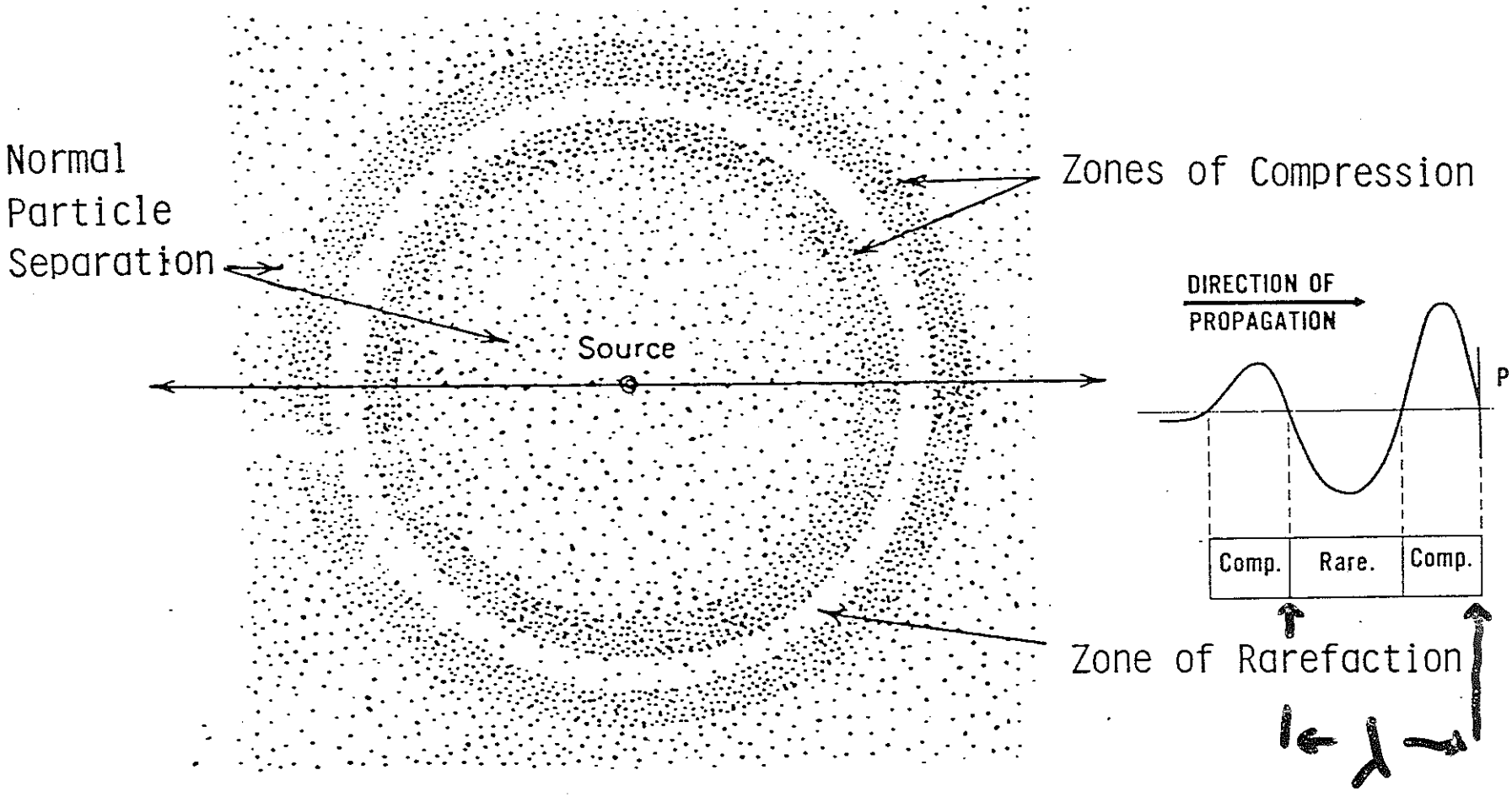
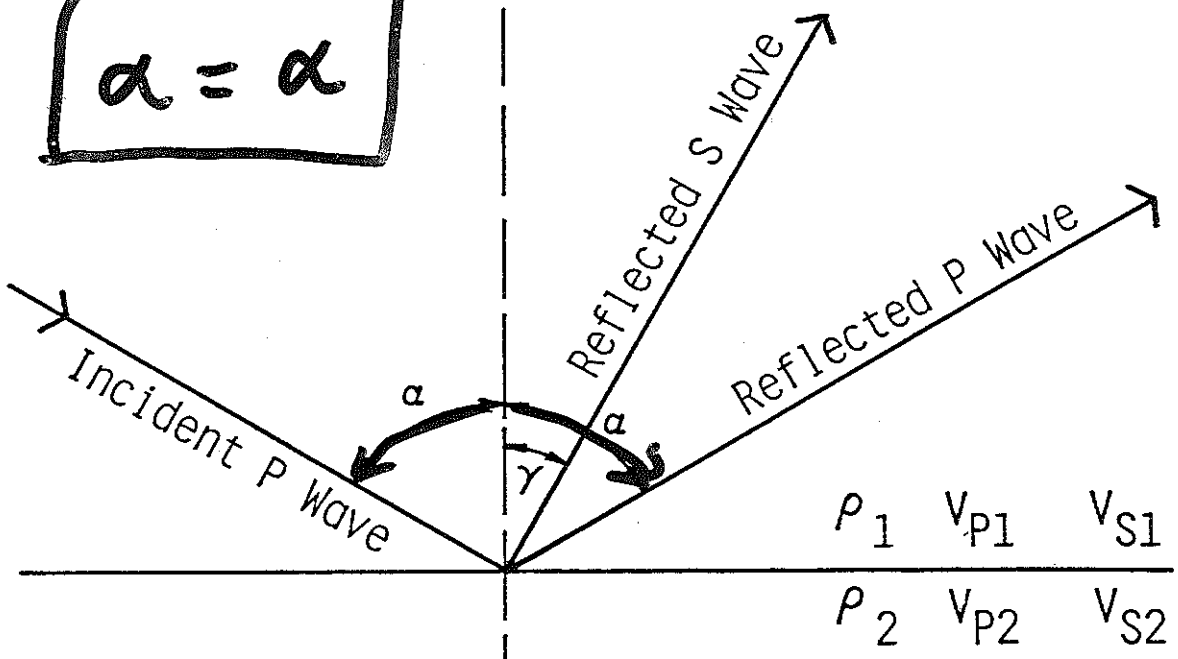
Compressional Waves



V = f lambda
Velocity equals Frequency times Wavelength
Memorize this!

Reflection

$$\boxed{\alpha = \alpha}$$



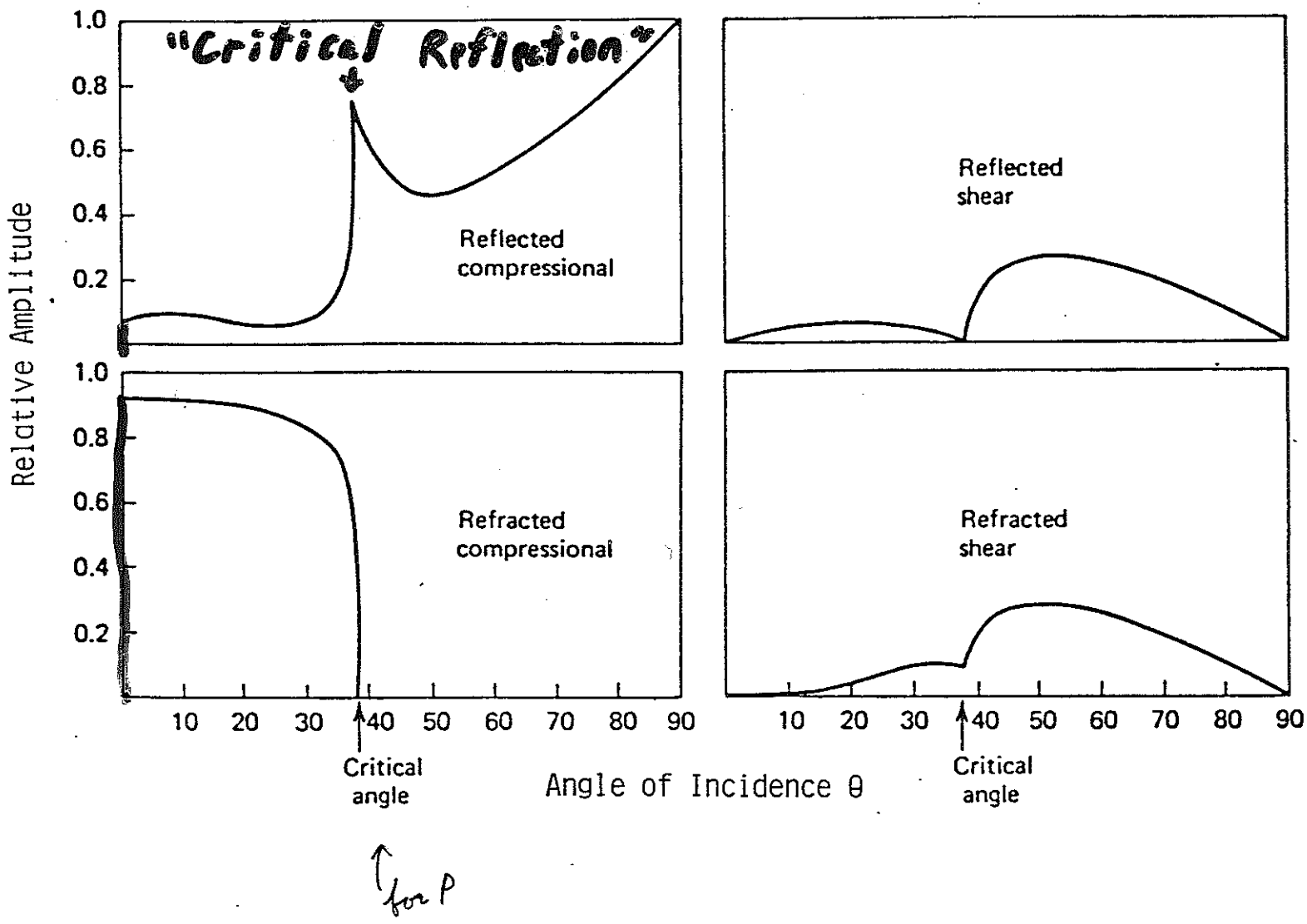
$$\sin \gamma = \frac{V_{S1}}{V_{P1}} \sin \alpha$$

When $\alpha = 0$

$$A_r = A_i \frac{\rho_2 V_{P2} - \rho_1 V_{P1}}{\rho_2 V_{P2} + \rho_1 V_{P1}}$$

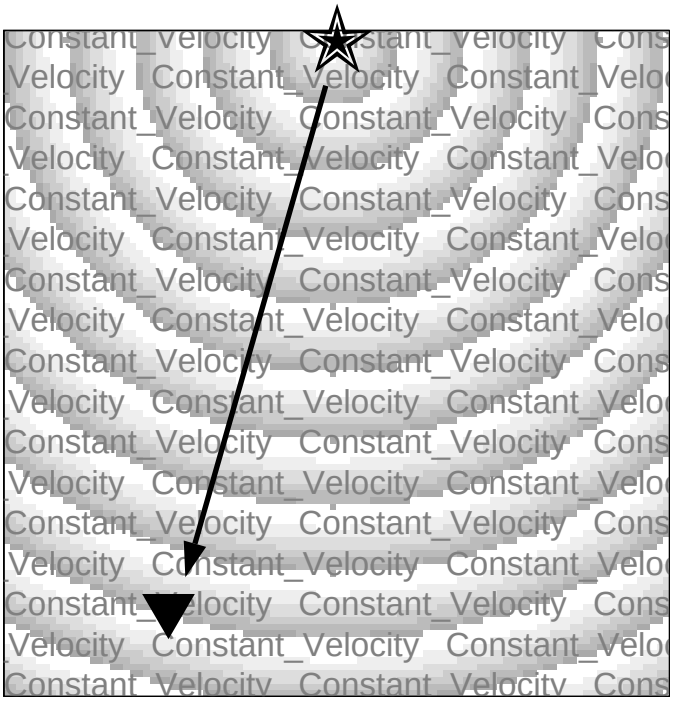
Amplitude Vs. Angle of Incidence

DIAGRAMMATIC



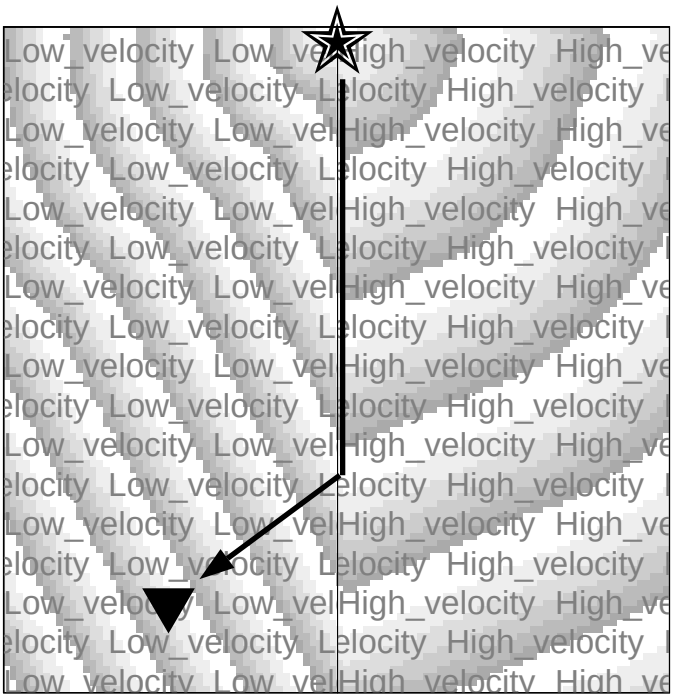
Fermat's Principle

"The wave path between any two points is the one along which the time of travel is the least of all possible paths" — the principle of ***Least Time***.



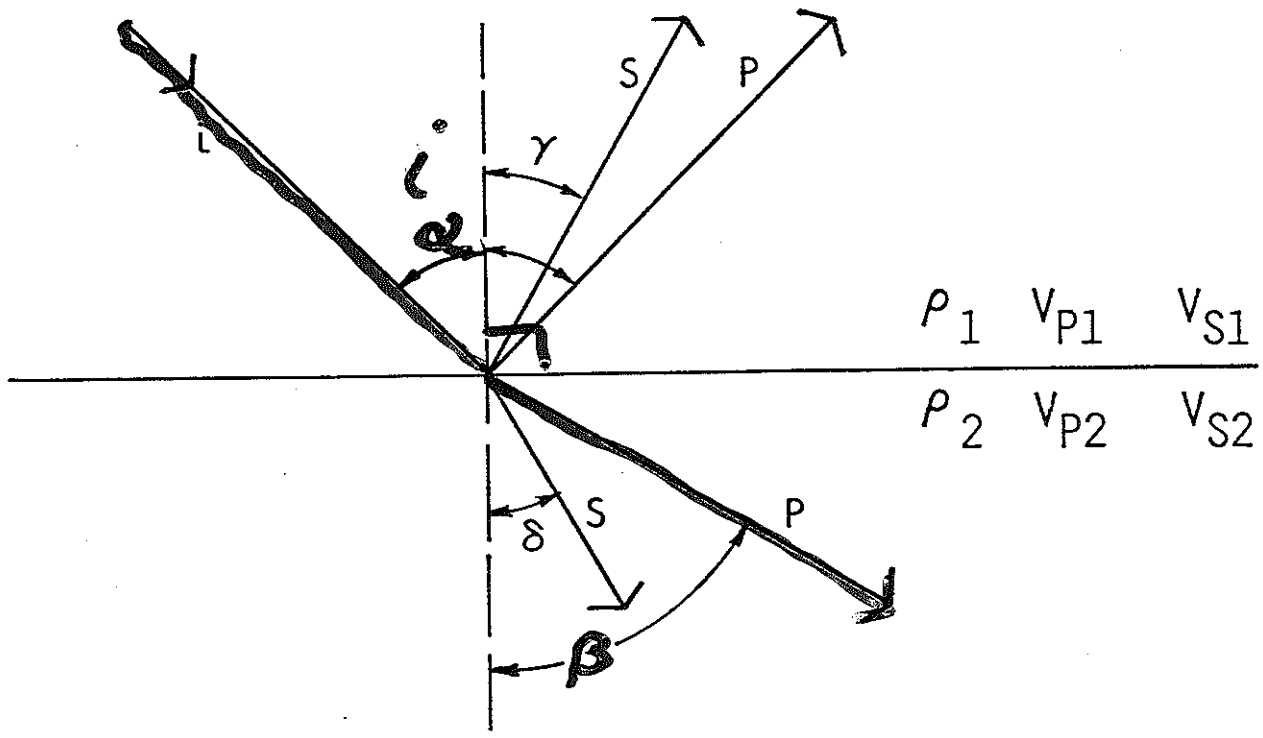
Ray paths are *perpendicular* to wavefronts.

Where velocity is *constant*, ray paths are *straight*.



Where velocity is *not* constant, ray paths will *bend*, to seek a high-velocity, least-time path.

Snell's Law

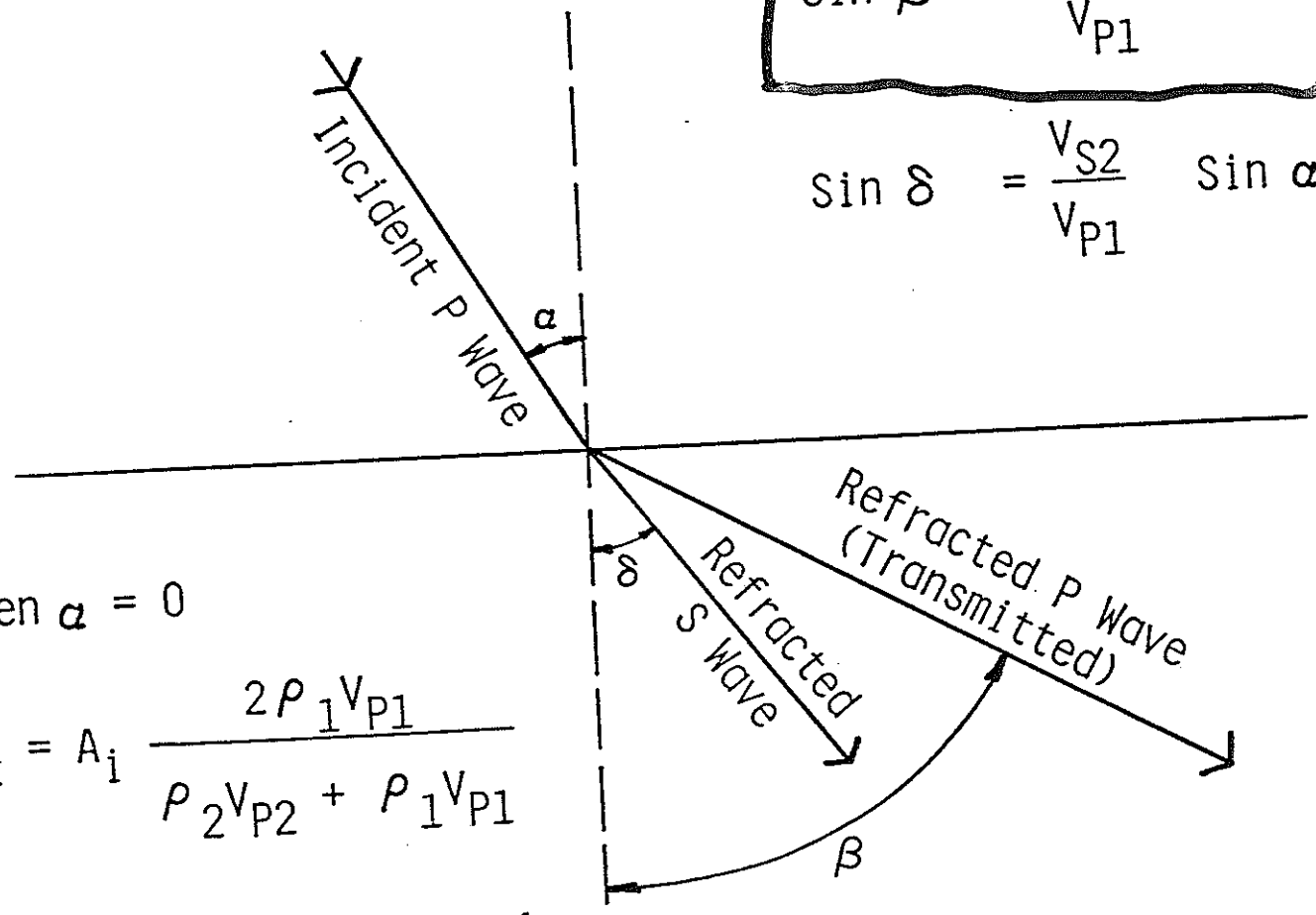


$$\frac{\sin i}{V_i} = \frac{\sin \alpha}{V_{P1}} = \frac{\sin \beta}{V_{P2}} = \frac{\sin \gamma}{V_{S1}} = \frac{\sin \delta}{V_{S2}} = \text{Const.}$$

Refraction

$$\boxed{\sin\beta = \frac{V_{P2}}{V_{P1}} \sin\alpha}$$

$$\sin\delta = \frac{V_{S2}}{V_{P1}} \sin\alpha$$

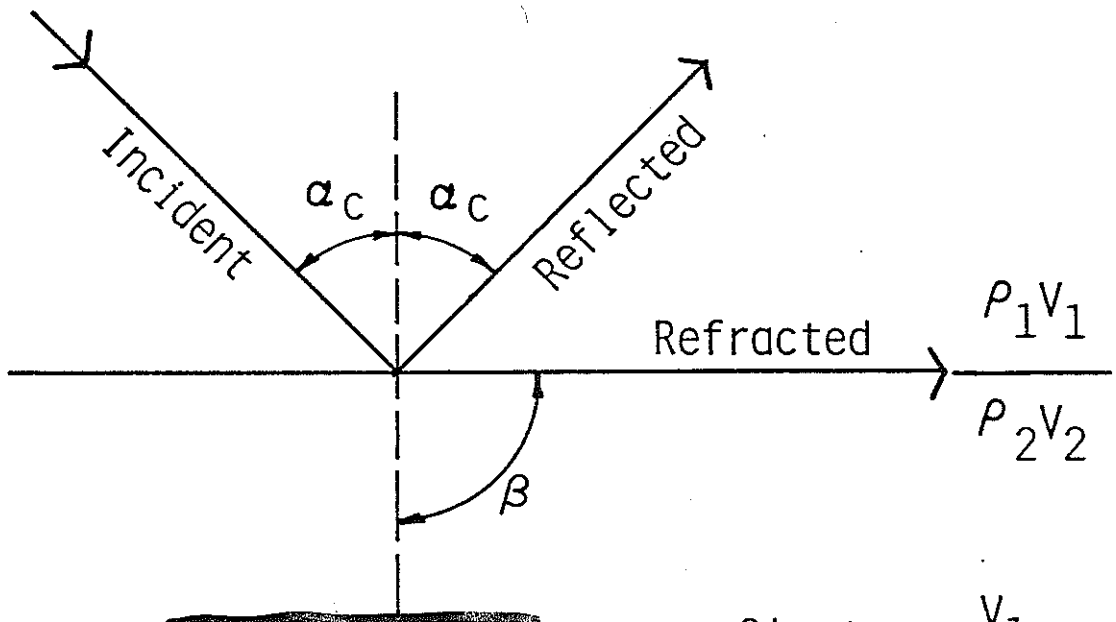


When $\alpha = 0$

$$A_t = A_i \frac{2\rho_1 V_{P1}}{\rho_2 V_{P2} + \rho_1 V_{P1}}$$

$\rho V = I$ = impedance

Critical Angle



$$\boxed{\sin \alpha_c = \frac{V_1}{V_2}}$$

$$\frac{\sin \alpha}{\sin 90^\circ} = \frac{V_1}{V_2}$$

What if $V_2 < V_1$?

Last Time

- Elastic Constants $\Delta P = k \frac{\Delta V}{V}$ $\quad V_p = \sqrt{\frac{k + \frac{4}{3}\mu}{\rho}}$
- Wave Equation, Propagation $V = f\lambda$

- Rock velocities:
  - ↑ water saturation
  - ↑ consolidation
  - ↓ weathering
  - ↓ fracturing

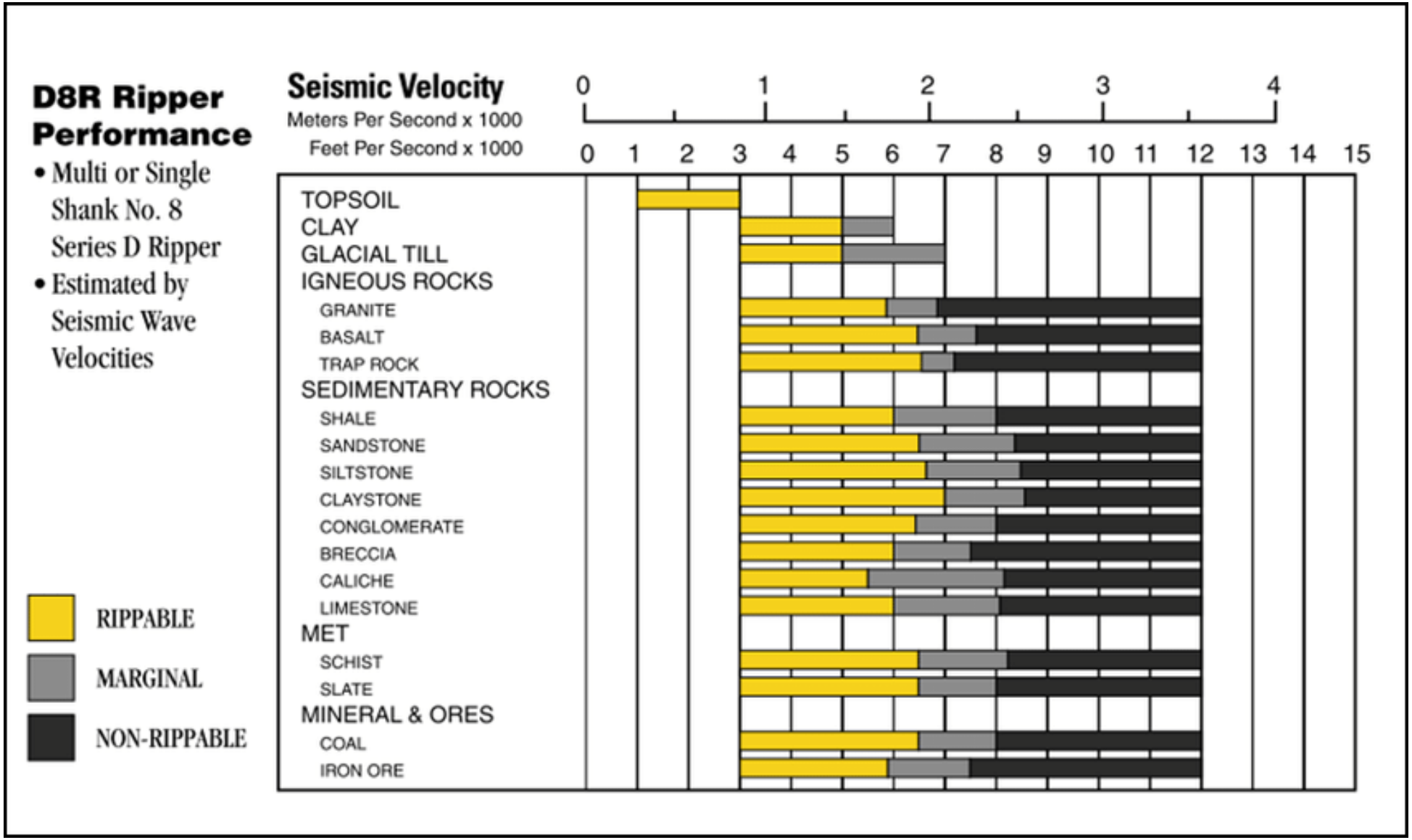
Also for Reference, V_p in km/s:

Unconsolidated Materials	
Weathered Layer	0.3 – 0.9
Soil	0.25 – 0.6
Alluvium	0.5 – 2.0
Calcretes	2.0 – 6.0
Clay	1.1 – 2.5
Sand, Unsat.	0.2 – 1.0
Sand, Saturated	0.8 – 2.2
Sand + Gravel	
Unsaturated	0.4 – 0.5
Saturated	0.5 – 1.5
Glacial Till	
Unsaturated	0.4 – 1.0
Saturated	1.7
Compacted	1.2 – 2.1

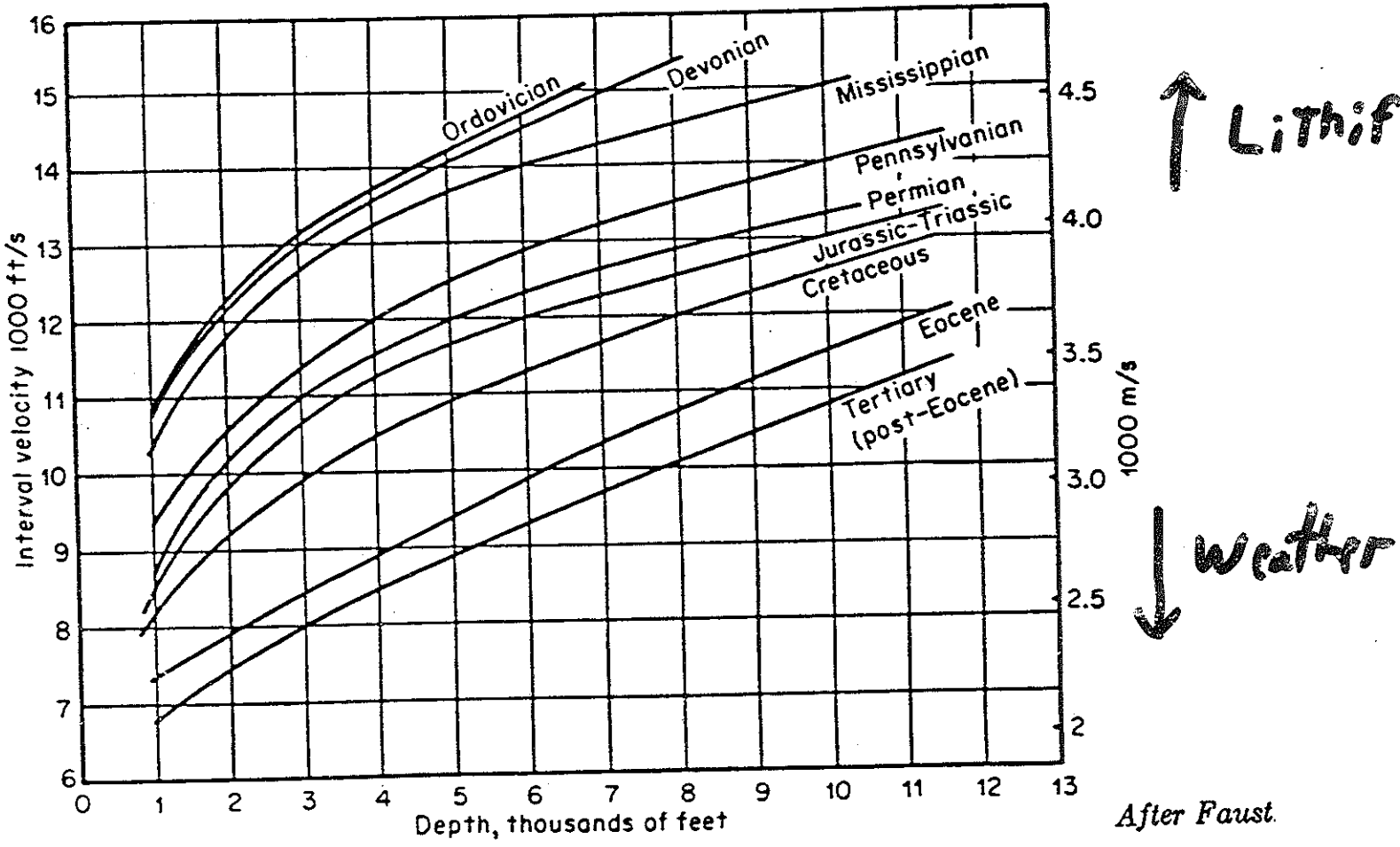
Consolidated Materials	
Granite	5.0 – 6.0
Basalt	5.4 – 6.4
Metamorphics	3.5 – 7.0
Sandstone + Shale	2.0 – 4.5
Limestone	2.0 – 6.0

Shear-Wave Velocities	
Type	V_s/V_p
Crystalline Rx	0.6 ($\sim 1/\sqrt{3}$)
Sedimentary Rx	0.5
Soils, Unconsol.	0.4
(Rules of Thumb)	

- Others: Fluids: $V_s \equiv 0$; Water 1.4 – 1.6; Air 0.32 – 0.34
  V_p for: Ice 5.0 – 6.0; Snow 0.2 – 5.0


D8R Ripper Performance
• Multi or Single Shank No. 8 Series D Ripper
• Estimated by Seismic Wave Velocities
Seismic Velocity
Meters Per Second x 1000
Feet Per Second x 1000
0 1 2 3 4
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
TOPSOIL
CLAY
GLACIAL TILL
IGNEOUS ROCKS
GRANITE
BASALT
TRAP ROCK
SEDIMENTARY ROCKS
SHALE
SANDSTONE
SILTSTONE
CLAYSTONE
CONGLOMERATE
BRECCIA
CALICHE
LIMESTONE
MET
SCHIST
SLATE
MINERAL & ORES
COAL
IRON ORE
RIPPABLE
MARGINAL
NON-RIPPABLE

V_P in Sandstones and Shales



After Faust.

$$V_P = K(ZT)^{\frac{1}{6}}$$

K = 125.3 (when Z is in feet) Z = Depth of burial (in feet)

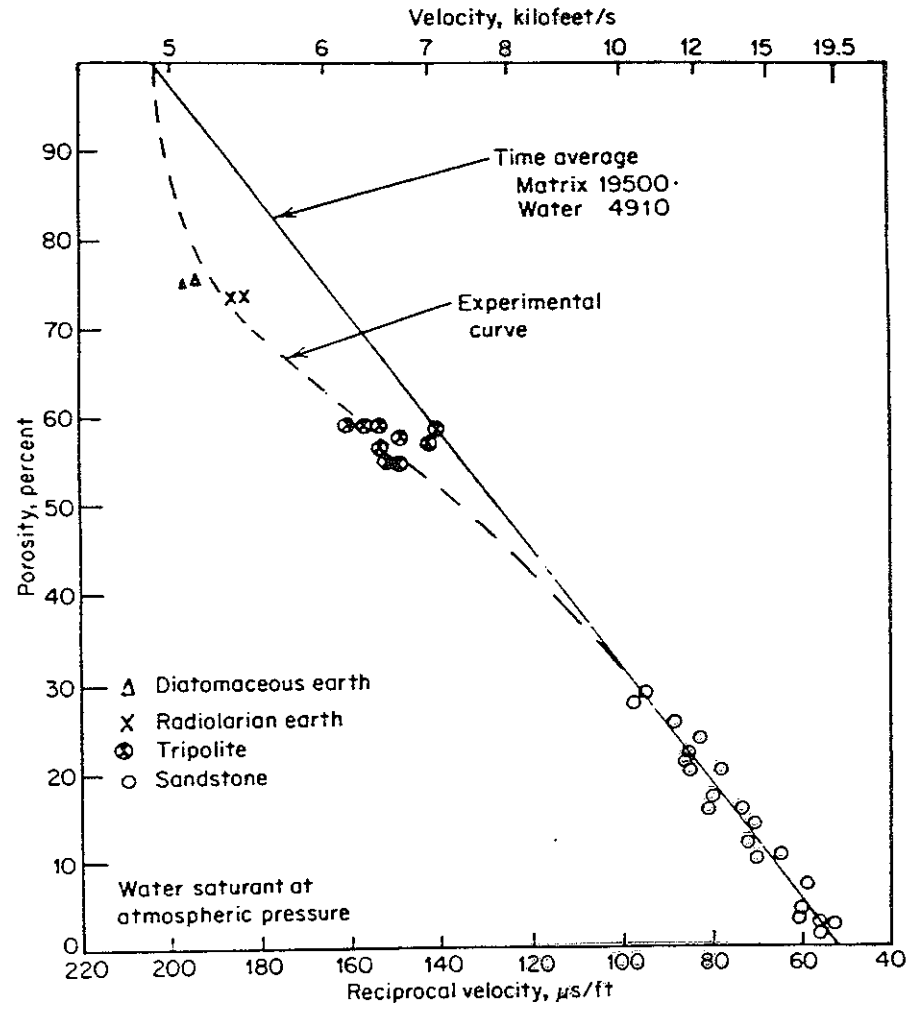
T = Age of formation (in years)

Velocity Versus Porosity

Wyllie
Time-Average
Relationship

$$\frac{1}{V} = \frac{\phi}{V_F} + \frac{1-\phi}{V_M}$$

where V = velocity in saturated rock
ϕ = fractional porosity
V_F = velocity of fluid in pore space
V_M = velocity of solid material making up rock matrix



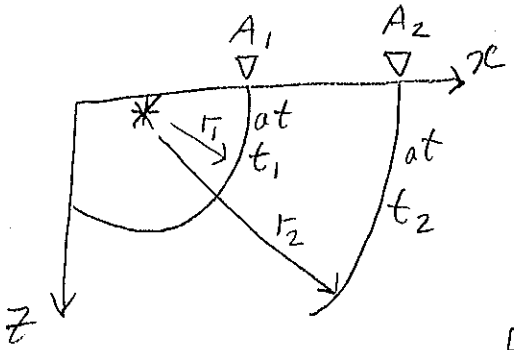
sonic logs

DOBRIN

Amplitude

We can only measure arrival times if they have enough energy to see through noise.

- Spherical Spreading



Energy is distributed over the surface area of the spherical wavefront.

$$\text{Area} = 4\pi r^2$$

$$\frac{E_2}{E_1} = \frac{4\pi r_1^2}{4\pi r_2^2} = \left(\frac{r_1}{r_2}\right)^2$$

Amplitude $= \sqrt{\text{Energy}}$, So $\dfrac{A_2}{A_1} = \dfrac{r_1}{r_2}$

→ At at distance r, $\boxed{A \propto \frac{1}{r}}$

- Absorption

Seismic waves lose energy to heat during each vibration cycle.

$$E = E_0 e^{-\frac{cfr}{VQ}}$$

c = constant
V = velocity
f = frequency
r = distance
$Q \equiv$ "Quality Factor"

High-frequency, low-velocity waves lose energy fastest

Example values of Q, the quality factor,

Dry Sand	10 - 50	Sedimentary Rock	50 - 1000
Alluvium	20 - 200	Igneous Rock	500 - 2000

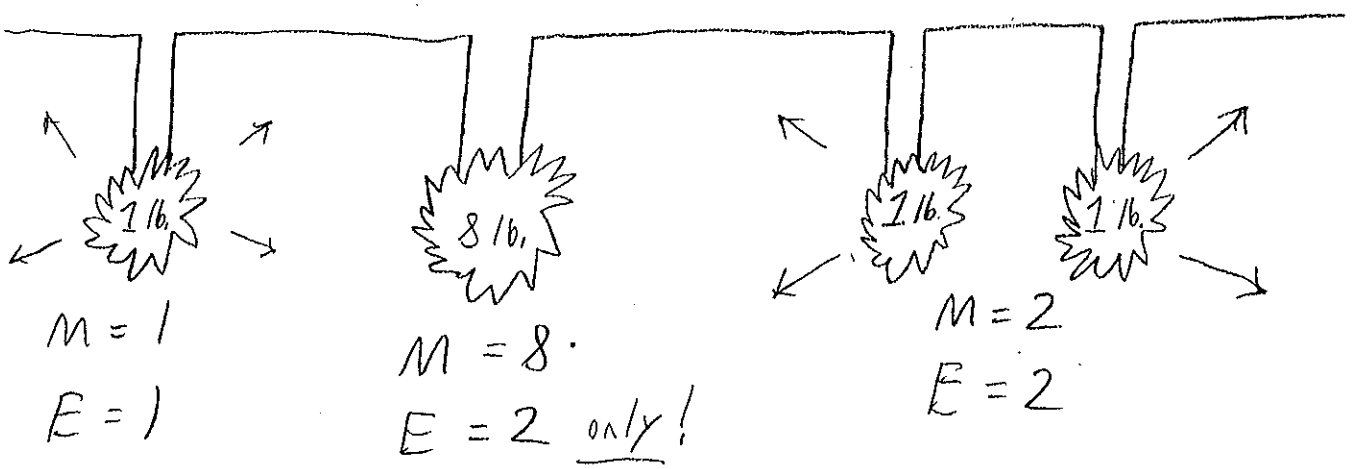
→ So high frequencies and shear waves die out very fast in <u>surficial</u> materials.

— When the <u>amplitude</u> of a seismic wave becomes very large, the absorption will increase drastically. This often occurs near seismic energy sources.

<u>Example:</u>

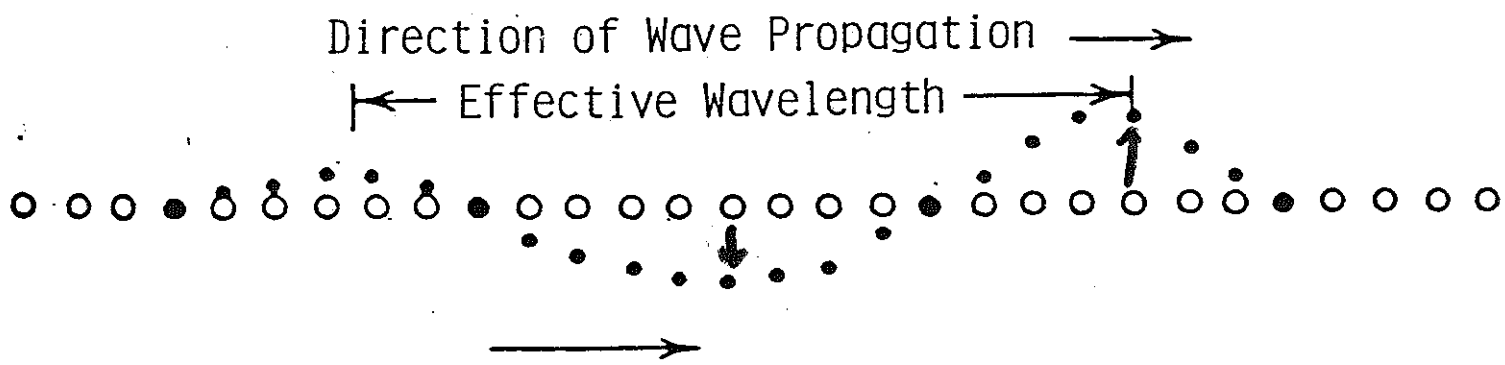
The seismic energy radiated by explosives placed at a single point increases only by a root of the increase in the mass of the explosive:

$$E \propto \sqrt[3]{M}$$



Putting more energy all at once at one place just produces more heat and breaks more rocks.

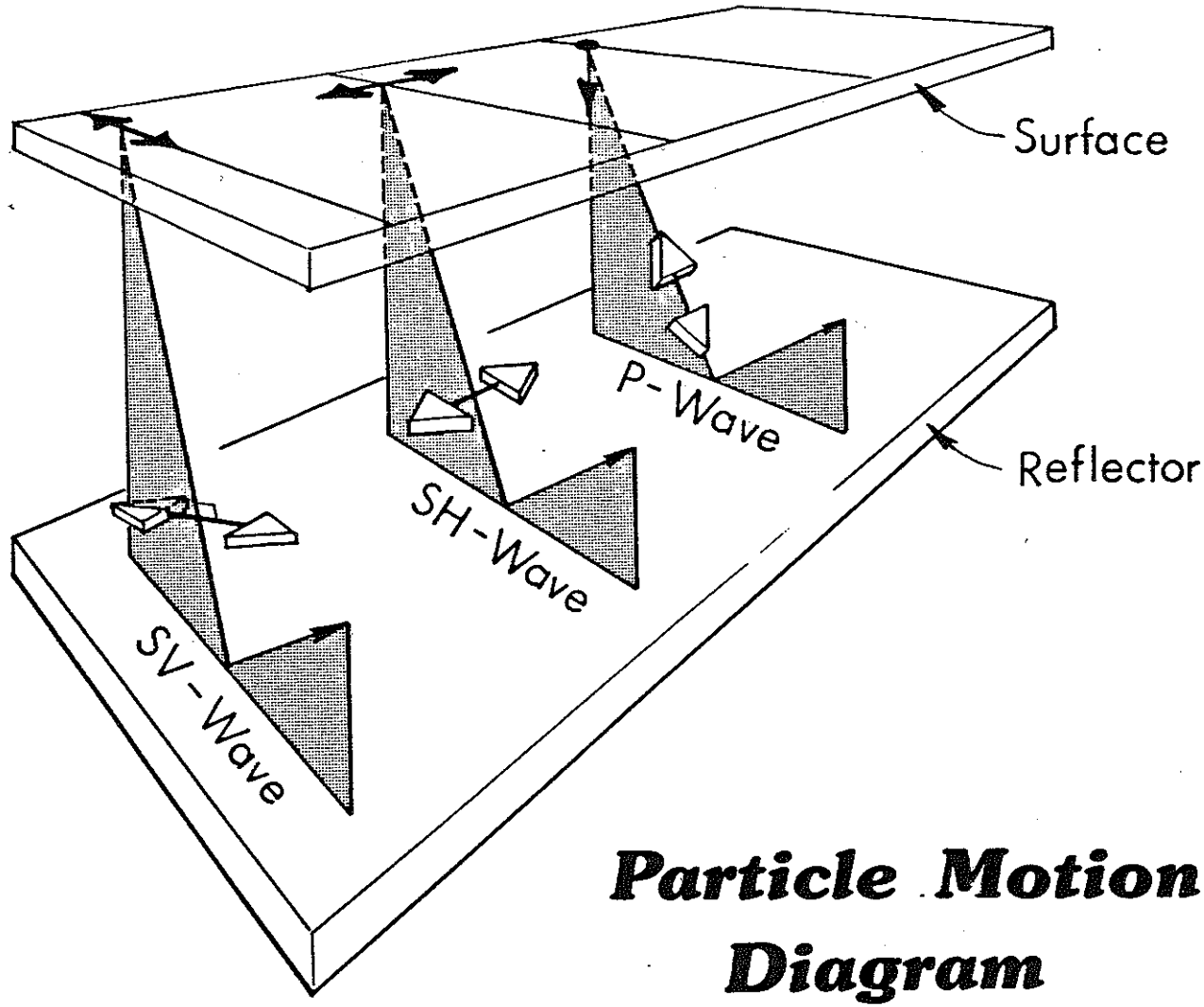
Shear Waves



o Normal Position of Particle

• Position During Passage of Shear Pulse

SH + SV

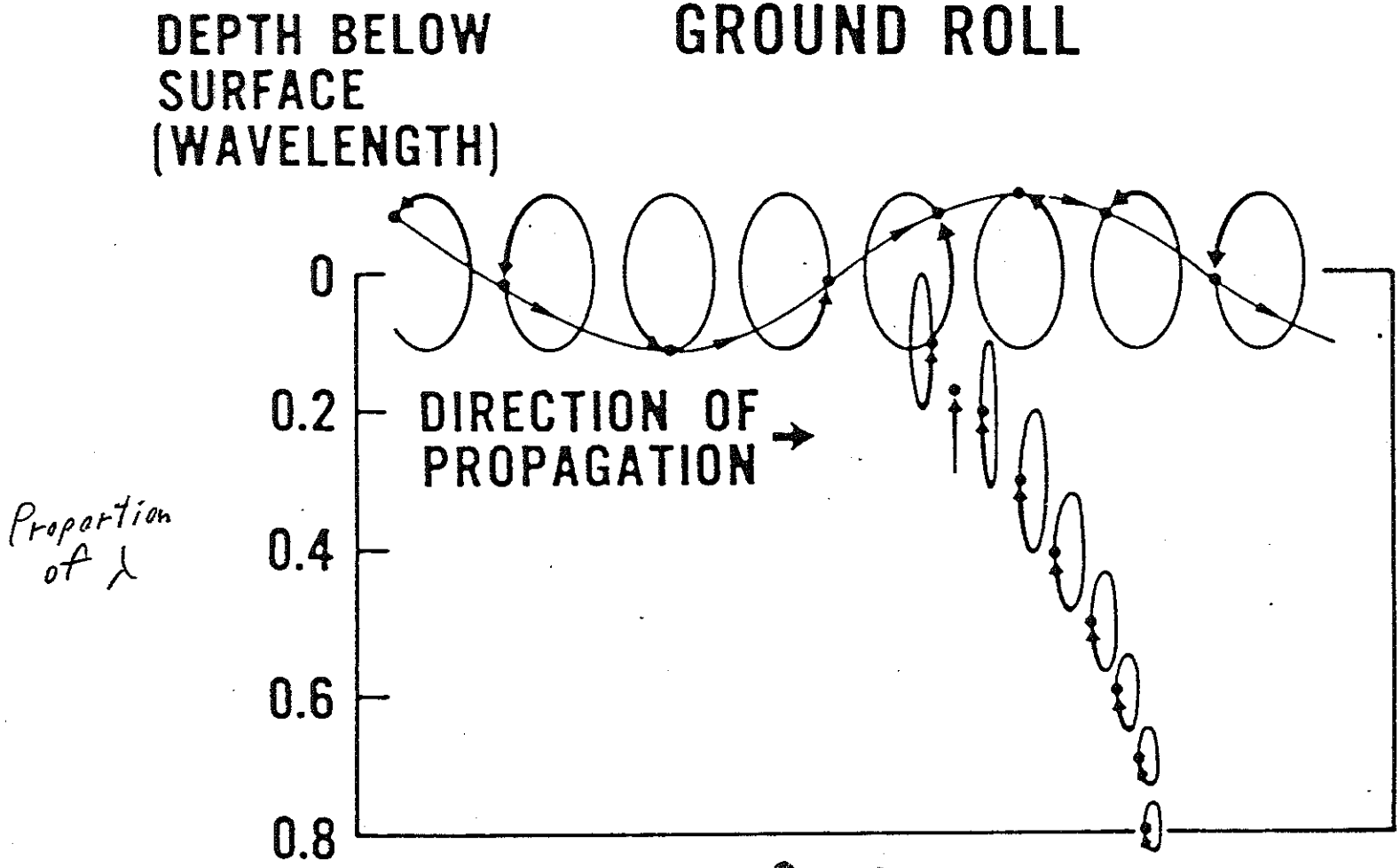


Particle Motion Diagram

Shear Vave work use SH — no conversions (assumptions?)

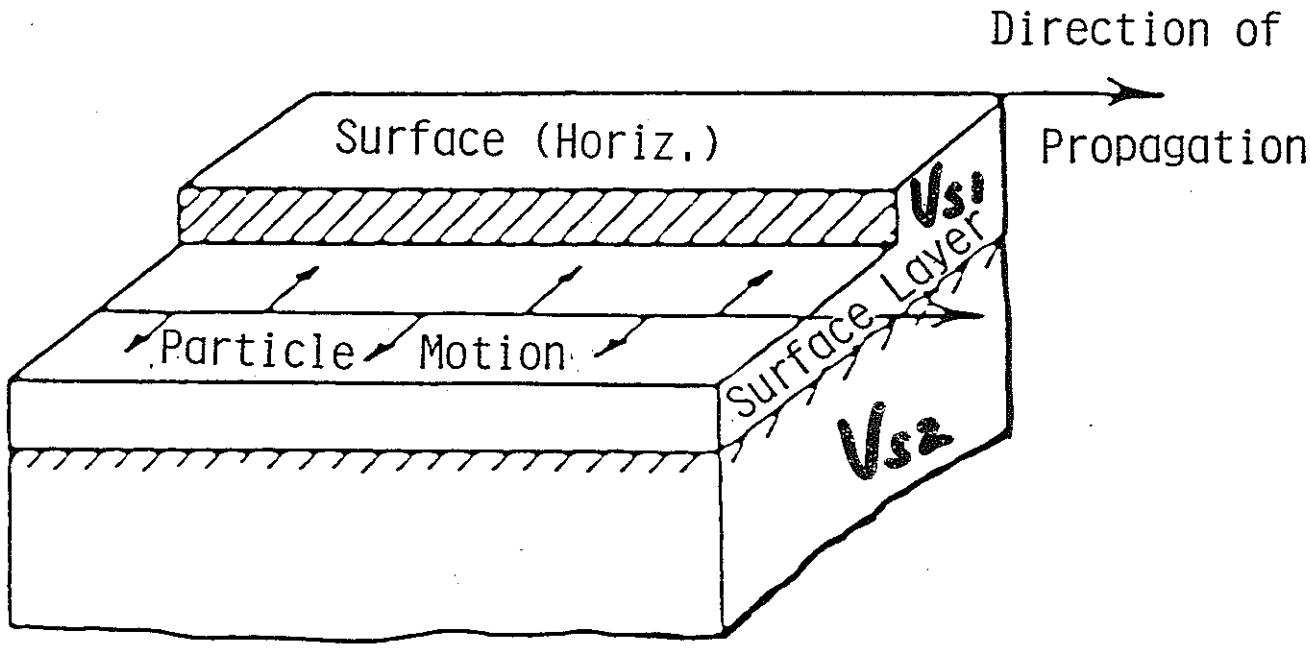
Rayleigh Waves

GROUND ROLL



$V_R \sim 0.9\ V_S$

Love Waves



$$V_L \sim V_{S1}$$

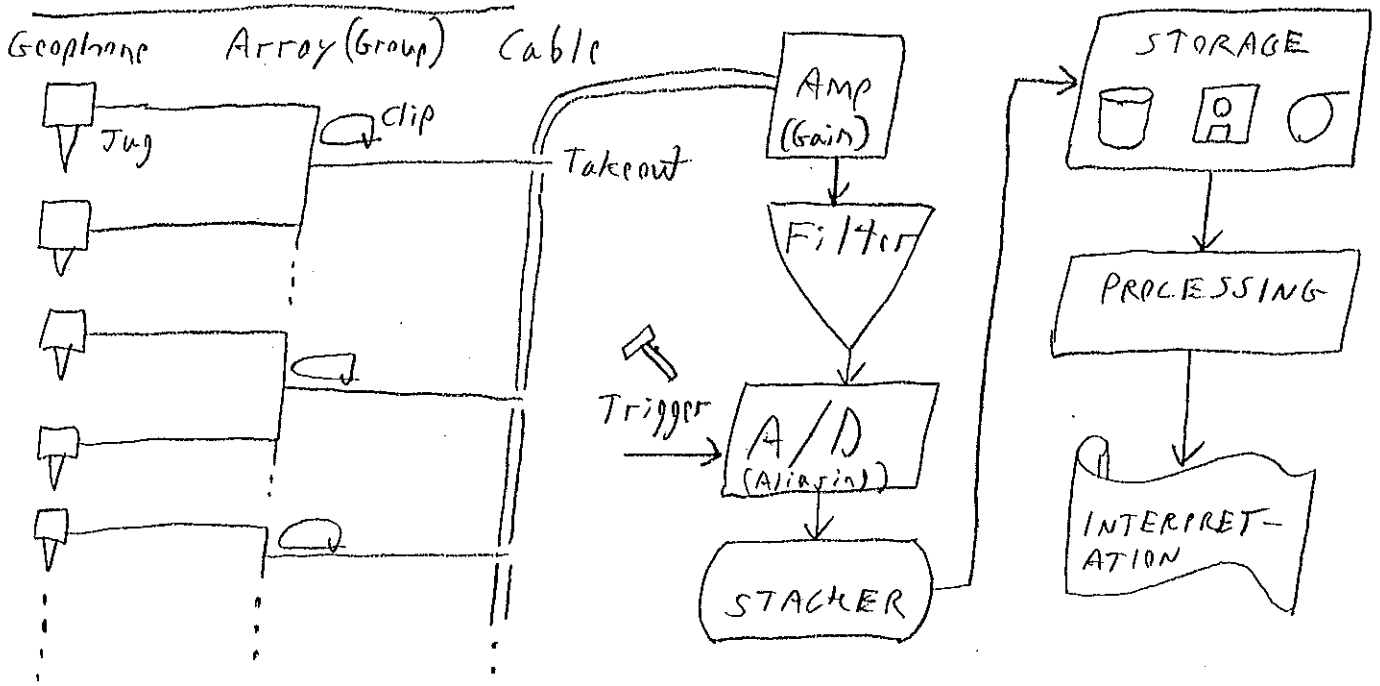
Seismic Sources

Passive	Advantages	Disadvantages
Earthquakes	Free. Source may be at location, depth otherwise inaccessible. Structure may be by-product of monitoring. High Energy – probe whole Earth.	Unreliable. Need many expensive receivers to establish time, dist. Low-frequency, lots of s-waves.
MICROTREMOR NOISE	**FREE. MORE IN URBAN AREAS.**	**LESS COHERENT NEED REC. ARRAY.**
Vibratory		
Vibroseis	Highly repeatable and linear. Inexpensive for many shot points. Frequency control, 10 – 150 Hz. To Moho.	Requires digital recording and extra processing step. Expensive Mobilization.
Mini-Sosie (Whacker)	50 – 200 Hz. Cheap. To 1 km depth. Arrays.	No real freq. control. Only 3 crews in U.S.
Marine + Borehole	50 – 1000 Hz	Low energy. Rental Expensive.
Explosive		
Shot hole	Energetic. High frequency. Repeatable. Fewer s- and air-waves. Place below low-vel. surface layer.	Permitting. Drilling costs. Non-linear. Blaster contract.
Air Blast	Cheap. High freq. Few s-waves. To 1 km.	Remote areas only. Big air wave. Linear.
Shotgun (Betsy)	Portable, reapeatable, High frequency. Cheap. To < 200 m only.	Low energy, time-consuming. Best off soft sediments, with digital recording.
Sparker	Marine + Borehole, high frequency.	Low energy, expensive mobilization.

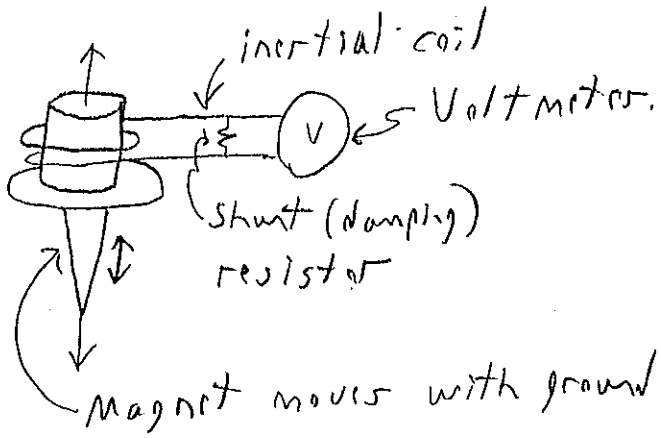
Impact	Advantages	Disadvantages
Land Air-Gun	Energetic, very repeatable and high-freq. To 1 km.	1 crew, expensive mobilization. Requires digital triggering and recording. Road only.
Weight-Drop (~75 kg)	Energetic, repeatable, to 1-2 km. No permitting.	Road only, constant maintenance. Not cheap enough. S-waves surface interference.
Sledge (~8 kg)	Most portable; repeatable. To 100 m. No permitting. High frequency. Cheapest.	Low energy, workers' comp. S-waves, air waves. Still require digital trig. + recording surface interference.

→ In survey planning, you must consider the material, labor, and service costs of large numbers of source "impacts."

Seismic Recorders



→ Geophones – large phonograph cartridges

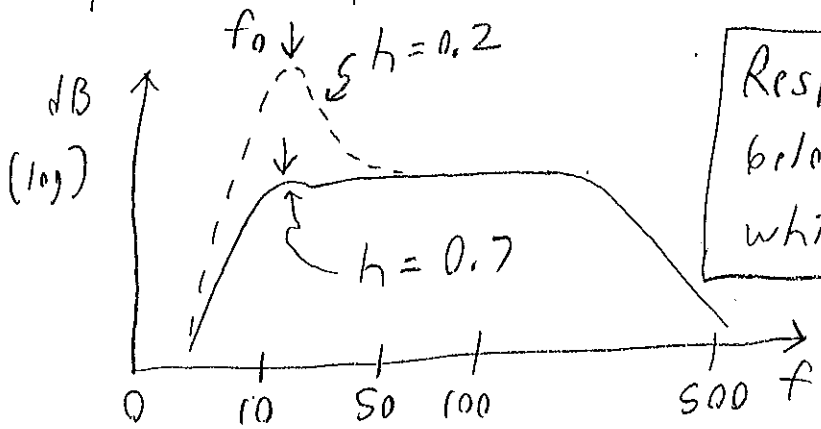


Voltage measured is proportional to ground particle velocity.

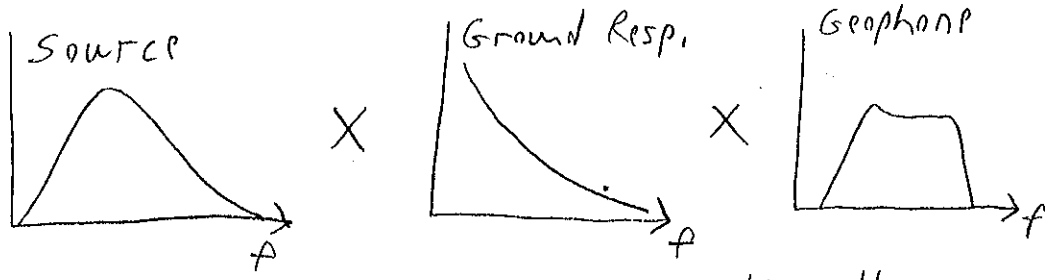
Two characteristics: $f_0 \equiv$ natural frequency

$h \equiv$ damping coefficient (% of critical damping)

Spectral response:



Response usually nil below f_0, flat for while above.



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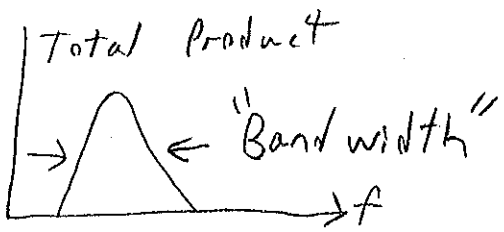


Usually,

→ Low limited by phones.

→ High limited by source and Q (distance)

<u>Filtering</u> → avoid data "saturation" by noise.

Anti-alias — only before A/D, steep cut at $\boxed{f_{NYQUIST}}$

$$f_{NYQ} = \frac{1}{2\,\Delta t} \qquad \Delta t = \text{sample interval.}$$

60 Hz Notch — AC power effect everywhere, usually before A/D.

Low-pass, High-pass, Band-pass — essential for work with inexpensive analog or <14-bit digital recorders. With modern recorders, may be applied only in later processing.
Set above ground roll, below air wave.

<u>Pointers</u>

- The Nyquist frequency is a hard theoretical limit. Stay 2–4 times away from it. (want 100 Hz data → f_{NYQ} = 200 Hz → sample above 400 Hz)
- Be prepared to spend substantial time testing instrument and survey parameters (Δt, T, filter stacks, Δx, X_{max}) with trial records.
- Watch records, re-evaluate, and re-test as survey progresses and conditions change.
- Per-hour contracts → good, expensive data
  Per-mile contracts → lousy, cheap data

Last Time (cont.)

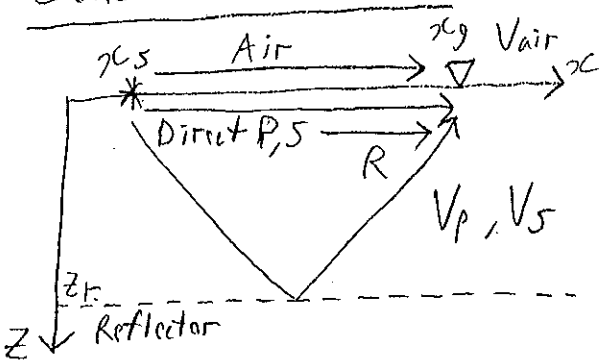
- Huygens Principle
- Fermat's Principle
- Snell's Law $\frac{\sin\theta_1}{V_1} = \frac{\sin\theta_2}{V_2} = \text{const.} = \frac{dt}{dx}$
- Refraction + Reflection, Critical Refraction $\sin\theta_c = \frac{V_1}{V_2}$ for $V_1 < V_2$

Time – Distance Plots

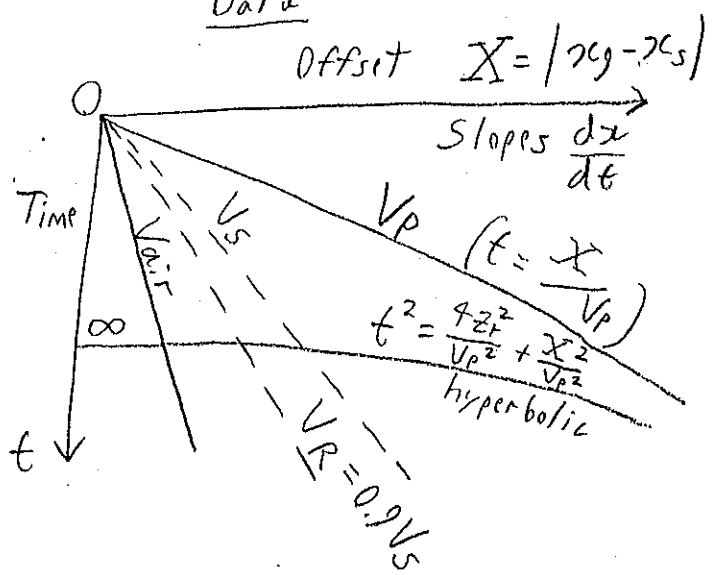
How to separate types (phases) of waves in our observations? Use their velocities:

Model

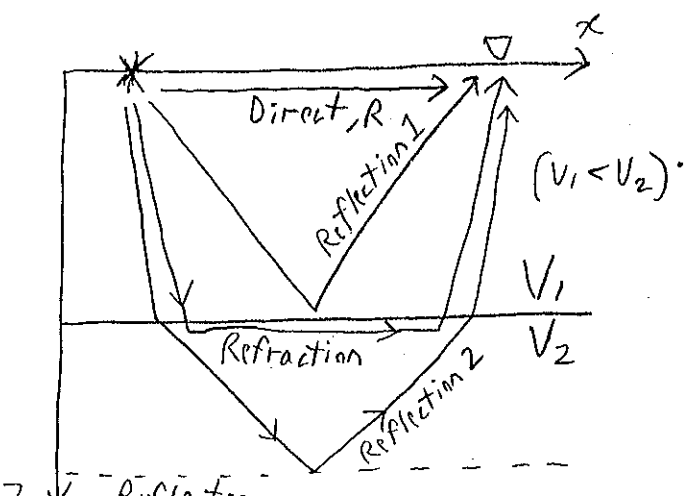
Constant Velocity

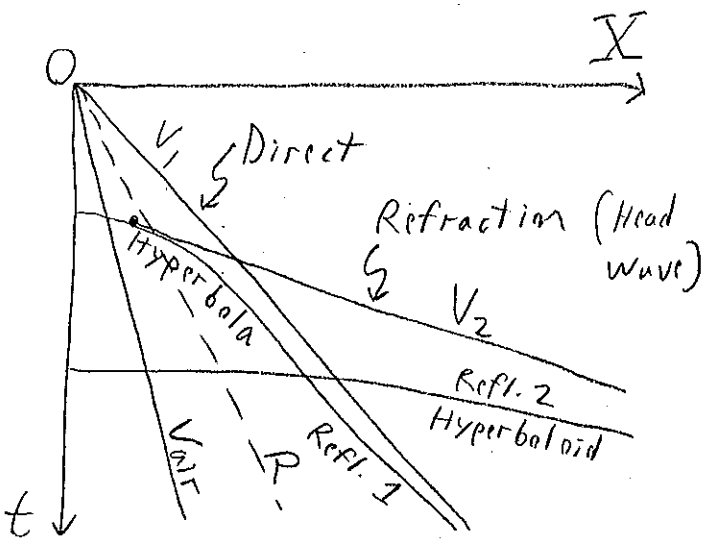


Data



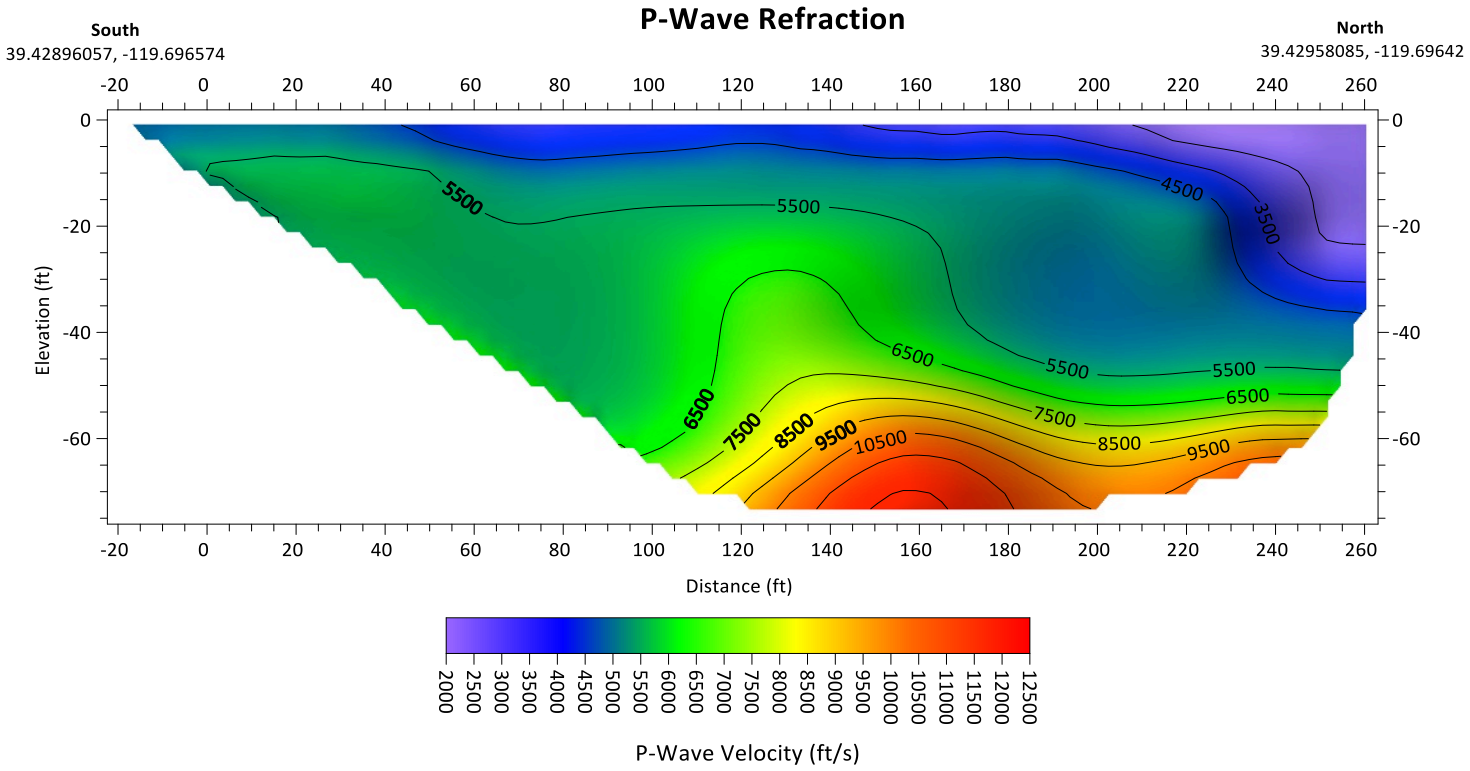
Layer-Over-Half-Space

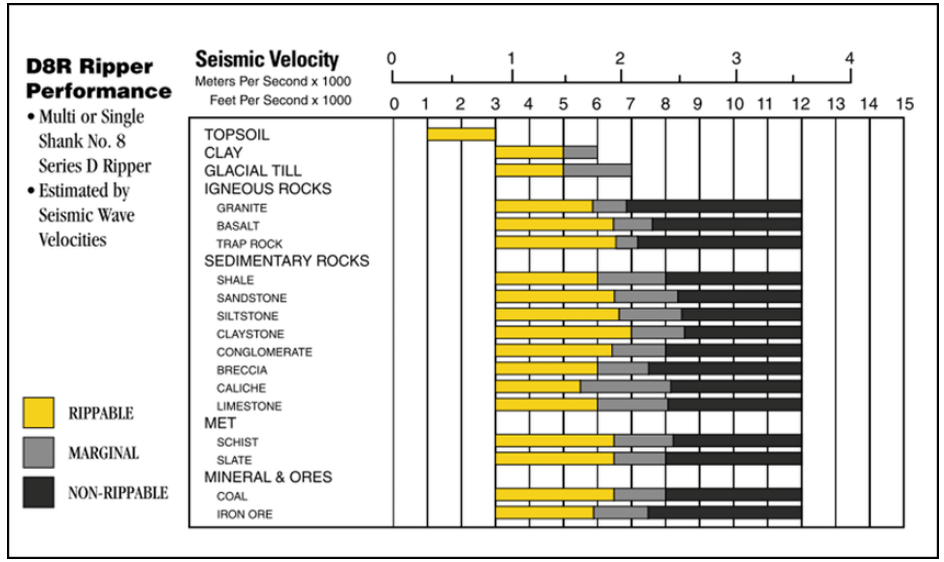




Application: Rippability

Summary: Below is a P-wave velocity output from the same project site. In addition to calculating IBC site class, we also determined Rippability. Under current rock Rippability guidelines, the site is considered Rippable to a 30-foot depth. Our study, along with a "*targeted*" boring program, allowed the client to complete a geotechnical report for their site, which was far more cost effective than the traditional extensive boring program.





Seismic Refraction

→ Direct estimation of rock velocities, often averaged over some distance.

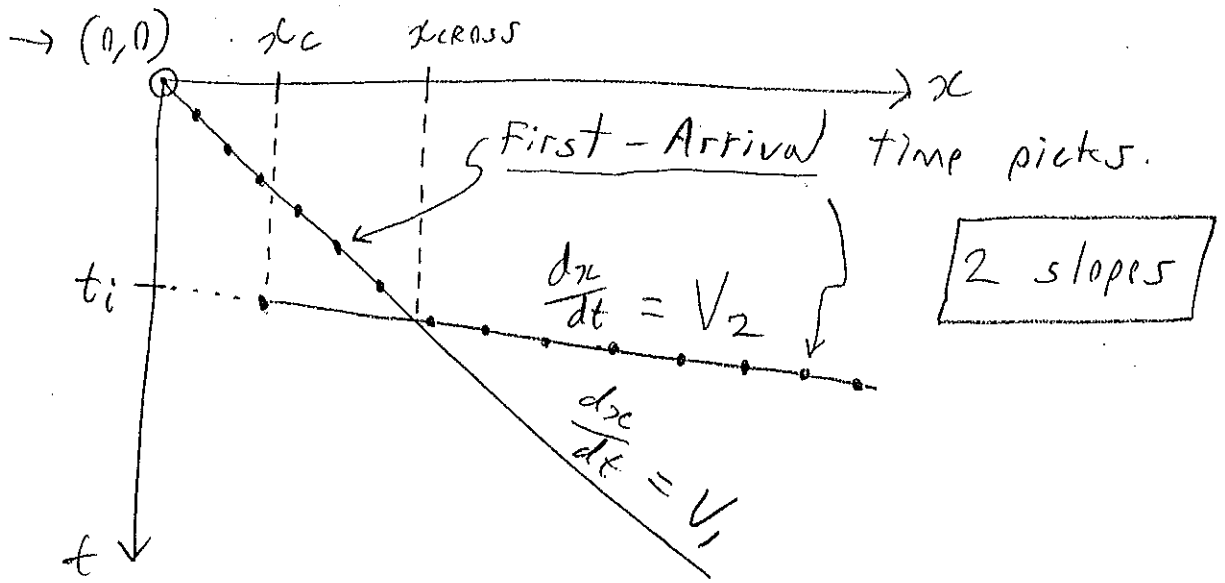
— Sounding: for depths to "Refractors" (velocity increases with depth).

— Profiling: for changes in z/v (depth/velocity) ratio above refractors, usually in terms of topography.

→ Most accurate for locating strong refractors such as bottom of unsaturated zone, floor of basin, or buried volcanics.

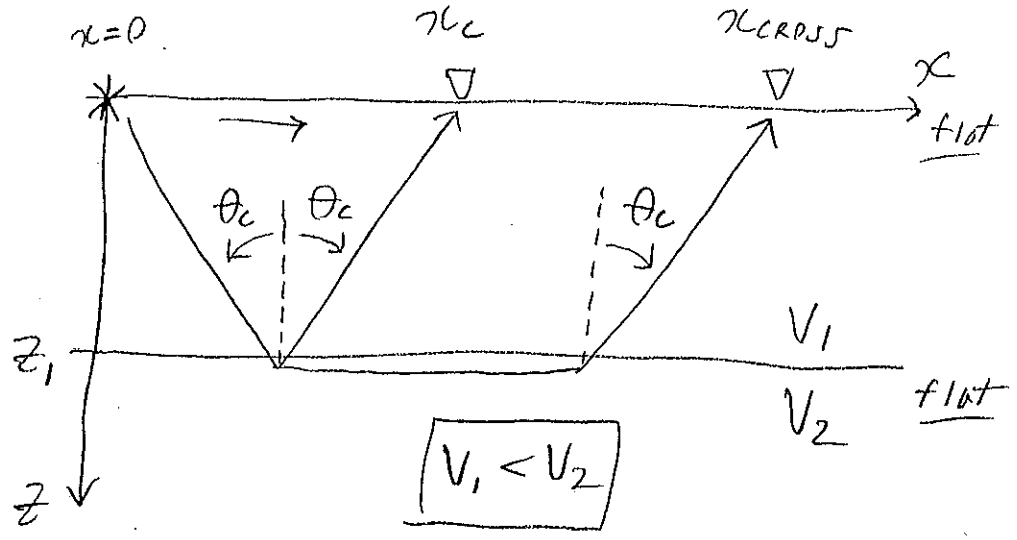
→ High sensitivity to fracturing, rippability.

Time-distance plot for layer over half-space:



$\boxed{\text{2 slopes}}$

Cross Section



$$\boxed{\theta_c = \sin^{-1}\frac{V_1}{V_2}} = \underline{\text{Critical Angle}}$$

V_1 = velocity of surface layer, from picks at $x < x_{cross}$. Find slope $V = \Delta x/\Delta t$, draw to (0,0).

V_2 = velocity of refractor, from $x > x_{cross}$

$$x_c = \frac{2z_1}{\sqrt{\left(\frac{V_2}{V_1}\right)^2 - 1}}$$ = Critical Distance, hard to observe, where refraction appears at surface.

$$t_i = 2z_1\frac{\sqrt{V_2^2 - V_1^2}}{V_2 V_1}$$ = Intercept Time, extend line from V_2 picks.

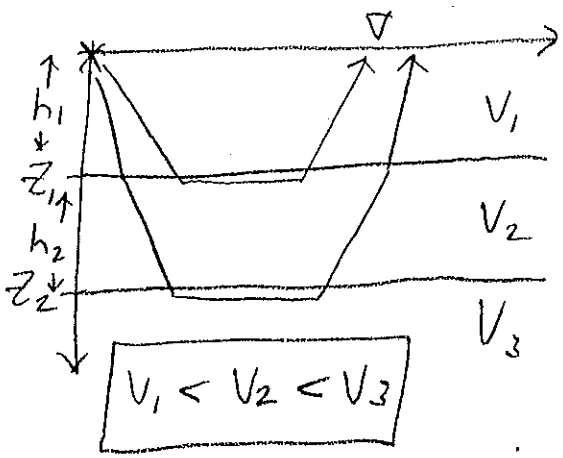
$$x_{cross} = 2z_1\sqrt{\frac{V_2 + V_1}{V_2 - V_1}}$$ = Crossover Distance, can observe where Refraction starts to arrive earlier than Direct wave.

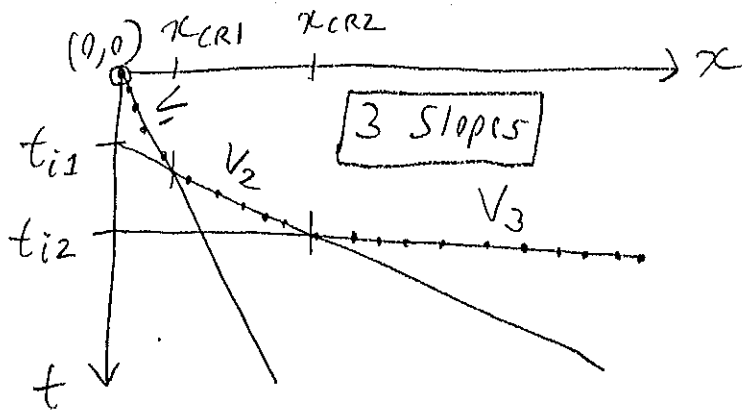
z_1 = Depth to refractor. Compute from:

<u>Intercept Time</u> or <u>Crossover Distance</u>

$$\boxed{z_1 = \frac{t_i}{2}\frac{V_2 V_1}{\sqrt{V_2^2 - V_1^2}}} \qquad \boxed{z_1 = \frac{x_{CROSS}}{2}\sqrt{\frac{V_2 - V_1}{V_2 + V_1}}}$$

<u>Two Refractors</u> – common as we often have a low-velocity surface layer.





<u>V_2 usually the most uncertain.</u>

→ V_1 may only come from closest phone — gives maximum.

z_1 = same, get from t_{i1} or x_{CR1} as above.

$z_2 = z_1 + h_2$

$$h_2 = \left[t_{i2} - \frac{2 z_1 \sqrt{V_3^2 - V_1^2}}{V_3 V_1} \right] \frac{V_3 V_2}{2\sqrt{V_3^2 - V_2^2}}$$

More computation → larger uncertainty.

Multiple Flat Layers

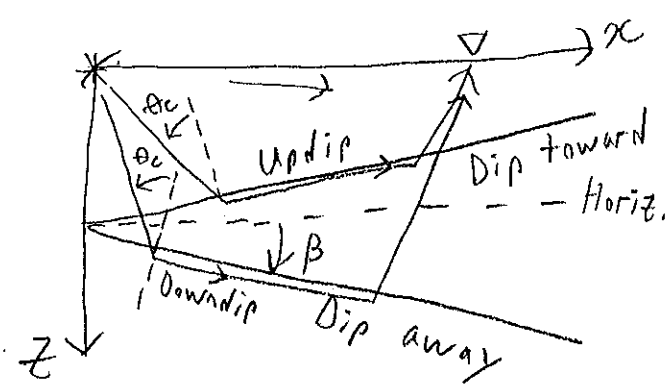
Computer programs will give the travel time for a wave critically refracted at the n-1 interface as:

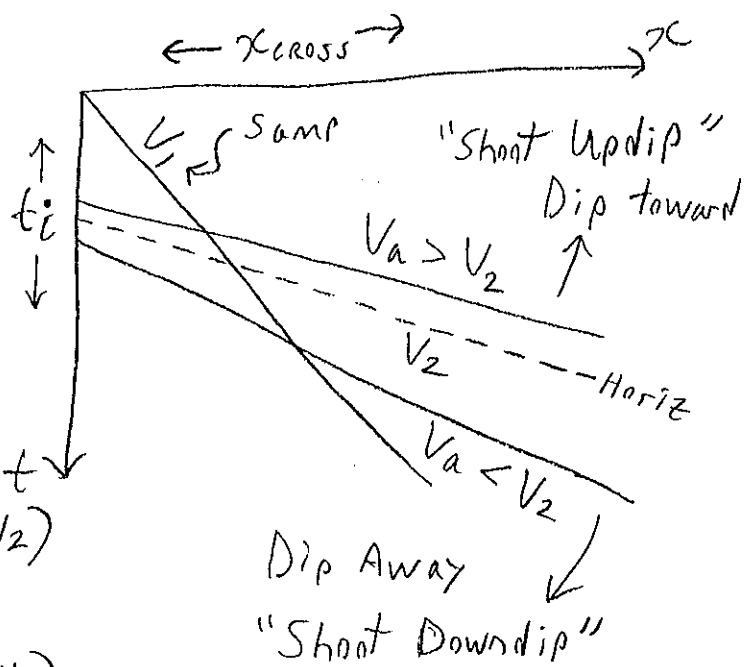
$$t_n = \frac{x}{V_n} + \frac{2}{V_n}\sum_{i=1}^{n-1} h_i \frac{\sqrt{V_n^2 - V_i^2}}{V_i}$$

h_i, V_i = thickness and velocity of layer i (1 is at surface).

Dipping Refractor

→ The slopes fitting 1st-arrival picks are not true velocities. They are apparent velocities that depend on true velocity and dip.





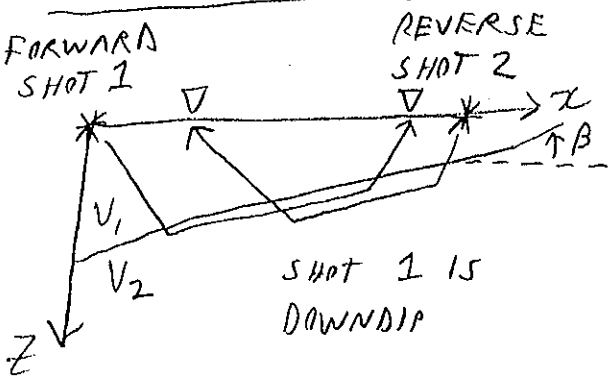
Updip: $V_a = \dfrac{V_1}{\sin(\theta_c - \beta)}$ $(> V_2)$

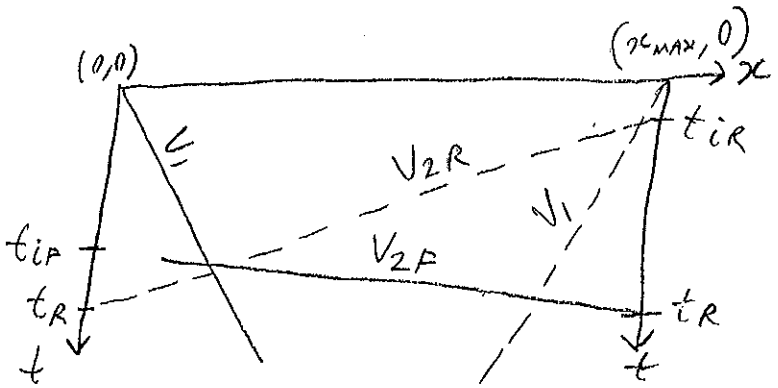
Downdip: $V_a = \dfrac{V_1}{\sin(\theta_c + \beta)}$ $(< V_2)$

All the equations above rely on having an accurate estimate of the refractor's velocity.

→ Perform additional experiments to constrain dip as well as velocity. <u>Always.</u>

<u>Reverse Shooting</u> — quickest solution





t_R = <u>Reciprocal time</u>. Source and receiver just switch position; seismic <u>reciprocity</u> says time is same.

V_{2F} = Forward apparent velocity.

V_{2R} = Reverse apparent velocity.

Forward shot is downdip, so here: $V_{2F} > V_{2R}$

and $t_{iF} > t_{iR}$

Now assign the <u>larger</u> of V_{2F}, V_{2R}, from the shot over the deeper part, to V_{2MAX}; the other to V_{2MIN}

<u>For $V_{2F} > V_{2R}$ (as above):</u>

$V_{2MAX} = V_{2F}$ $\quad t_{iMAX} = t_{iF}$

$V_{2MIN} = V_{2R}$ $\quad t_{iMIN} = t_{iR}$

<u>For $V_{2F} = V_{2R}$:</u>

$V_{2MAX} = V_{2R}$ $\quad t_{iMAX} = t_{iR}$

$V_{2MIN} = V_{2F}$ $\quad t_{iMIN} = t_{iF}$

Now to compute <u>velocity</u>, <u>depth</u>, and <u>dip</u>:

First get critical angle:

$$\theta_c = \frac{1}{2}\left[\sin^{-1}\left(\frac{V_1}{V_{2MAX}}\right) + \sin^{-1}\left(\frac{V_1}{V_{2MIN}}\right)\right]$$

since $\sin\theta_c = \frac{V_1}{V_2}$ $\boxed{V_2 = \frac{V_1}{\sin\theta_c}}$

Dip:

$$\boxed{\beta = \frac{1}{2}\left[\sin^{-1}\left(\frac{V_1}{V_{2MIN}}\right) - \sin^{-1}\left(\frac{V_1}{V_{2MAX}}\right)\right]}$$

Depths:

$$\boxed{Z_{MAX} = \frac{V_1\, t_{iMAX}}{2\cos\theta_c \cos\beta} \qquad Z_{MIN} = \frac{V_1\, t_{iMIN}}{2\cos\theta_c \cos\beta}}$$

Survey Design

→ <u>Maximum Offset</u> Use a modeling program on a set of extreme (end-member) layered models to get the crossover distance for the deepest refractor you want to find. Rule of Thumb: usually 3 to 6 times the depth desired.

→ <u>Source</u> Consult local experts for experience on what sources will reach your maximum offset. Local conditions can vastly change source costs.

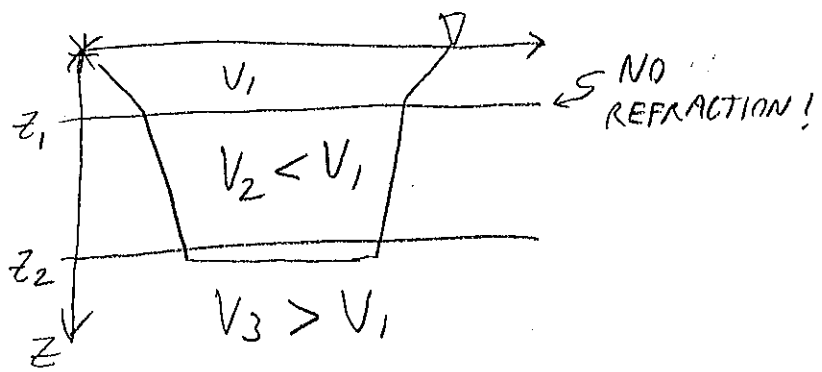
→ <u>Reverse</u> Plan at least a test shot, forward shot, and reversal. You may need more.

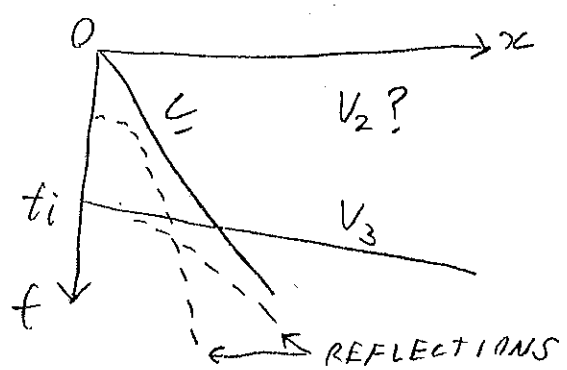
Complications

Situations that violate assumptions made above, leading to erroneous results:

- Low-Velocity Layer
- Thin Layer
- Lateral Velocity Change
- Faults

Low-Velocity Layer (Hidden Zone 1)

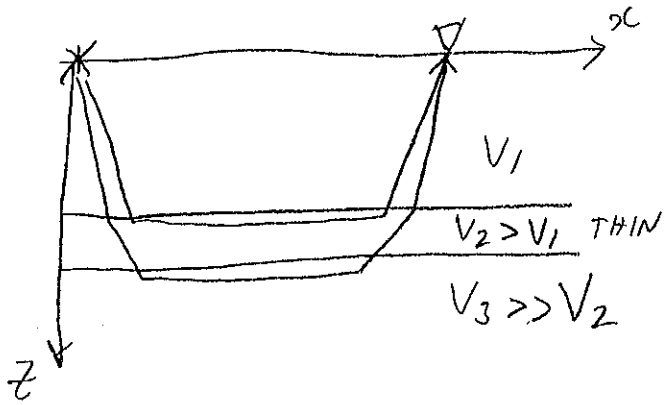


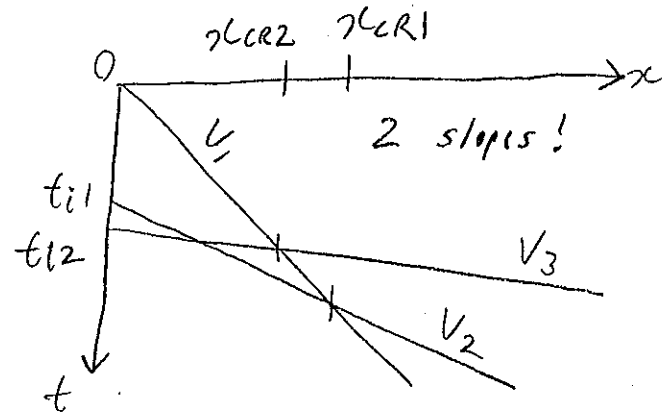


Problem: Only 2 slopes in time-distance plot, so you use equations for 2 velocities. But the waves spend extra time in the low-velocity layer, so t_i (and x_{cross}) is too large. You will get a false depth that is larger than the true depth z_2. OK for drilling plans, disaster for aquifer assessment. An Equivalence problem.

Solution: None for refraction alone. Can control if you collect good enough data to measure the reflections. LV layers often low density, so mismatch between gravity and refraction is a clue. The V_1/V_2 transition is often a big resistivity change. Borehole data are best.

Thin Layer (Hidden Zone 2)



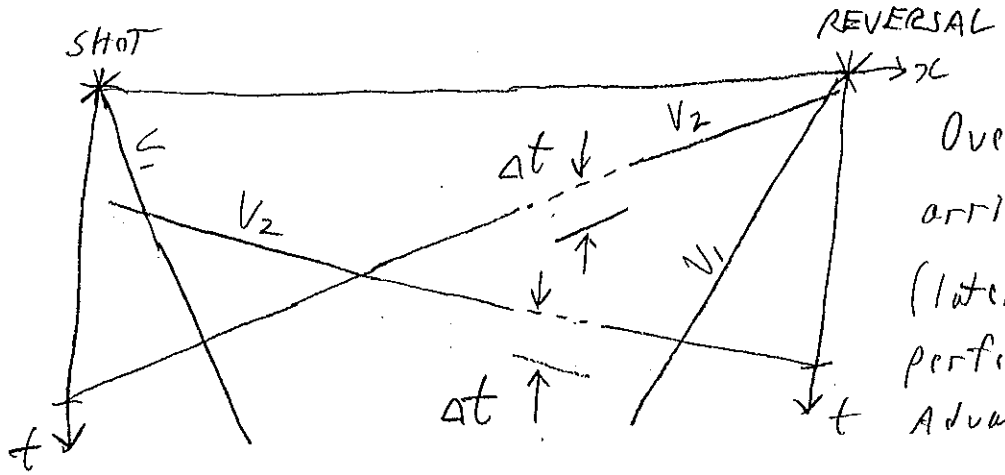


Problem: Layer with V_2 is thin, t_{i1} is too close to t_{i2} and $x_{CR1} > x_{CR2}$, so V_2 never becomes first arrival. Waves travel faster than V_1 in the layer, so t_{i2} is too little and the false depth for V_3 is smaller than the real depth. A minimum depth.

Solution: None for refraction alone. With any idea about the velocity between V_1 and V_3, put that slope at the first arrivals at x_{CR2}, giving a maximum depth. (See Lab) This situation is very common, and only known stratigraphy, drilling, or maybe reflection can control it.

Lateral Changes

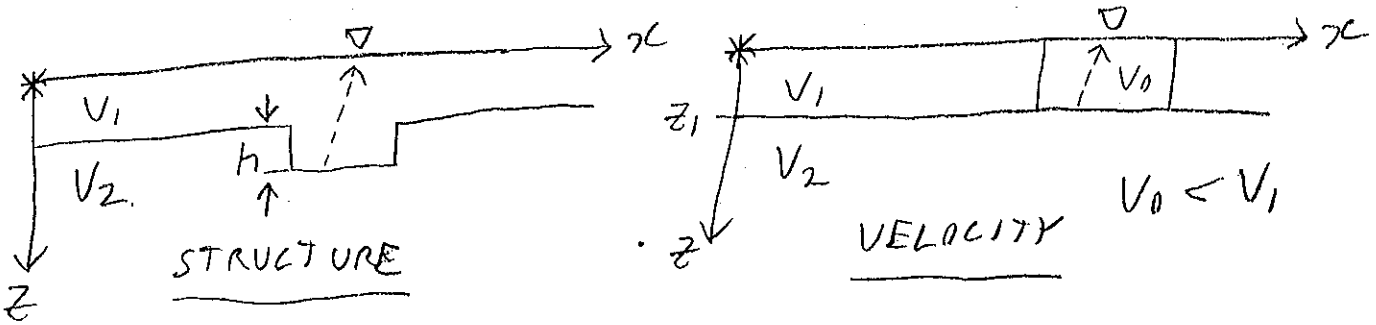
Of course you cannot pass perfect straight lines through all your first-arrival time picks.



Over some range, arrivals are <u>delayed</u> (later) relative to the perfect straight lines.
Advance = − Delay

→ Advances or Delays occurring at the same geophones for both forward and reverse shots, with about the same amount, are likely from a lateral change <u>directly under</u> the delayed phones. (Thicker V_1 layers will shift forward + reverse delays away from each other more.)

Basic Ambiguity: Is it structure or is it velocity?



Both situations can produce the <u>same data</u>.

A <u>delay</u> produces a <u>positive</u> Δt.

An <u>advance</u> produces a <u>negative</u> Δt. (A negative delay)

→ To analyze a delay you have to assume structure or velocity, or a combination, hopefully based on corroborating data.

- Structure in refractors, like fault blocks, may have a potential-field signature.

- Near-surface velocity changes may be associated with easy-to-detect resistivity variations.

→ Get a definitive answer by putting your source in the delay area, to see if V_1 really changes there.

- With <u>structure</u> assumption, compute <u>deflection</u> h (maybe a fault throw):

$$\boxed{h = \frac{\Delta t \cos\theta_c}{\frac{1}{V_1} - \frac{1}{V_2}}}$$ (for zero refractor dip)

- With <u>velocity</u> assumption, compute the new velocity V_D (a vertical average, like V_1):

$$\boxed{\frac{1}{V_D} = \frac{\Delta t \cos\theta_c}{Z_1} + \frac{1}{V_1}}$$ (again zero dip)

Last Time

Seismic Refraction → velocities, refractors

- pick first-arrival times
- measure: $V_1, V_2, \ldots, t_i, x_{cross}$
- compute: true V, dip, depths

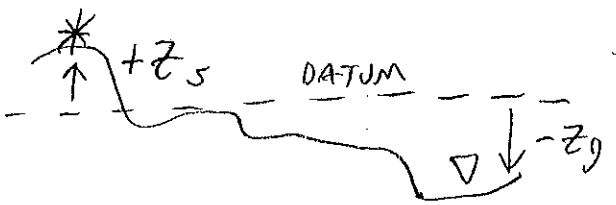
→ Need reversal to constrain dip

→ X_{max} is 3-6 times deepest depth.

- Complications:
  1) Low-velocity layer → reflection, gravity, EM
     z_{max}
  2) Thin Layer → reflection
     z_{min}, get z_{max} from V_2 guess
  3) Lateral Changes → structure/velocity ambiguity
     more shots, reflection

- Methods for Irregular Refractors → **0) OPTIMIZATION - SEE LAB**
  1) Delay-Time - assumes no hidden layers, veloc. changes
     dip < 20°
  2) Wavefront Construction - tedious
  3) Generalized Reciprocal Method (GRM) → tomography
     Propose model, trace ray paths by full application of Snell's Law, adjust until times fit.
     >20° dip, thin layers controllable.

- **Elevation Corrections** - if shot + receivers not all at same level.



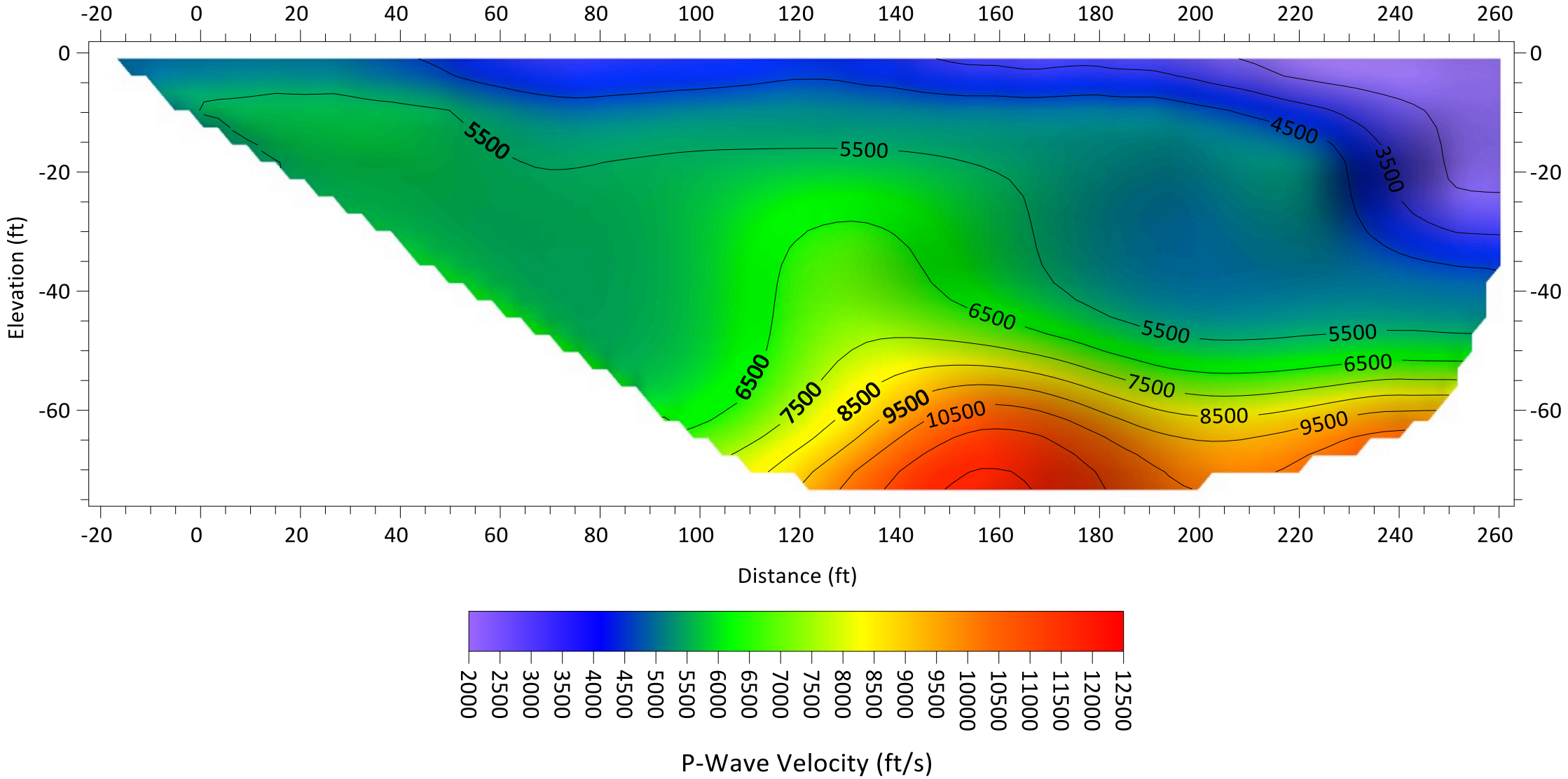
$$\Delta t = (z_s + z_g)\frac{\sqrt{V_2^2 - V_1^2}}{V_2 V_1}$$

SUBTRACT from geophones' time.

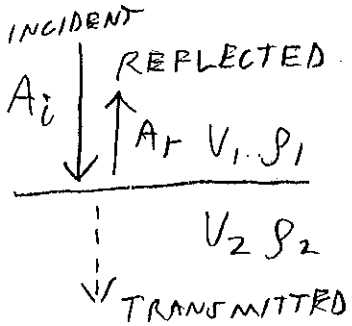
P-Wave Refraction

South
39.42896057, -119.696574

North
39.42958085, -119.69642



SEISMIC REFLECTION Principles



Where elastic properties of rock change, there can be a reflection.

$$A_r = A_i \frac{\rho_2 V_2 - \rho_1 V_1}{\rho_2 V_2 + \rho_1 V_1} \propto \frac{\Delta I}{I}$$

Integrative Methods → data from average property over large surrounding area.

→ Great Accuracy, low precision

Examples: Potential-field, galvanic, EM amplitude Refraction methods

Derivative Methods → data from change in property over small area.

→ Low Accuracy, great precision

Examples: EM phases (Radar), Reflection

Reflection Methods give detail, find differences between nearby points → Imaging, picture of cross section.

– Implies higher Cost, as only more data can give more detail on more points.

Advantages of Reflection

1) PRECISION — can get sufficiently-detailed image to "discover" unsuspected structures

2) PENETRATION — often the cheapest method to sound to any particular depth.

3) STRATIGRAPHY — Images nearly as detailed as trench logs allow complete sedimentological analysis.

4) LOCATION — Most accurate placement of structures, disturbances.

Disadvantages

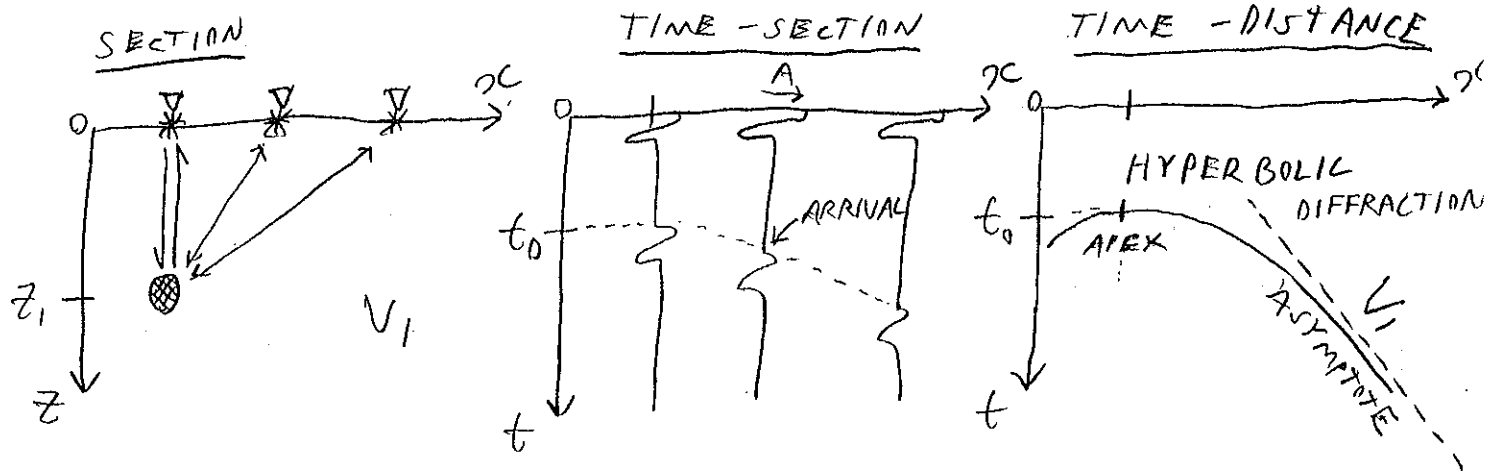
1) Cost — only cheap over water, labor-intensive

2) Near-Surface Heterogeneity — where strong will make getting good data <u>much</u> too expensive

3) Misunderstanding — Detailed time-sections fool geologists into thinking they are looking at a cross-section.

— Reflection is usually the last method (before drilling) called upon to confirm and resolve conflicting results of cheaper methods.

<u>Pure Profiling</u> – put source, geophone in same spot, then move and repeat



Directly over the structure:

$$\boxed{t_0 = 2\frac{z_1}{V_1}}$$

measure time, assume velocity, get depth.

Elsewhere:

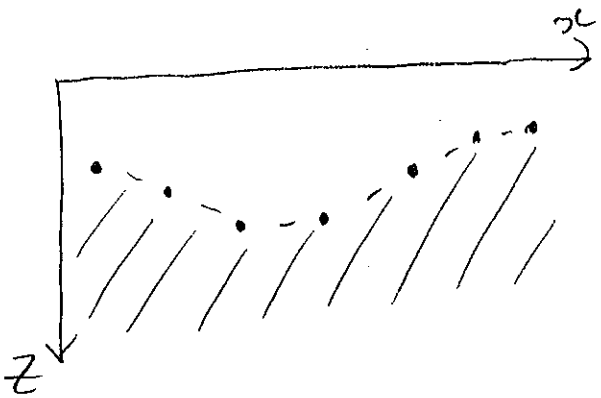
$$\boxed{t^2 = 4\frac{z_1^2}{V_1^2} + 4\frac{x^2}{V_1^2} = t_0^2 + 4\frac{x^2}{V_1^2}}$$

Collect times at two distances, and you can solve two equations for the two unknowns: depth z_1 and velocity V_1!

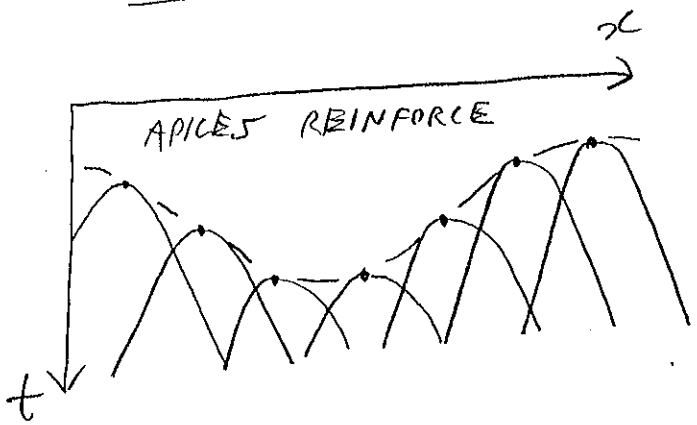
→ Point diffractor yields hyperbola centered above it, asymptotic to straight line of slope V_1.

— According to Huygens Principle, any continuous structure (a <u>reflector</u>) can be approximated by a large number of point <u>diffractors</u>.

<u>SECTION</u>

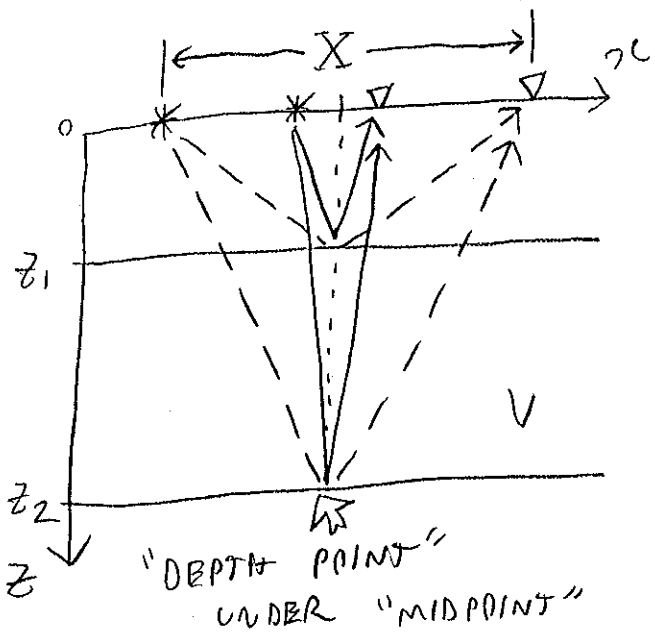


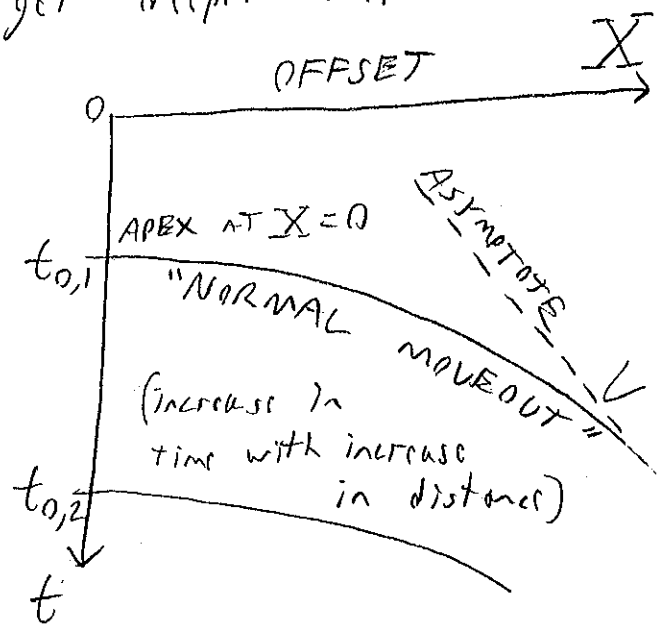
<u>TIME – DISTANCE</u>



— Pure Profiling is usually called a <u>zero-offset</u> survey, <u>single-channel</u> survey, or <u>echo-sounding</u>.

<u>Pure Sounding</u> — increase array aperture over one spot to get deeper depths:





NORMAL MOVEOUT (NMO)

For this geometry also:

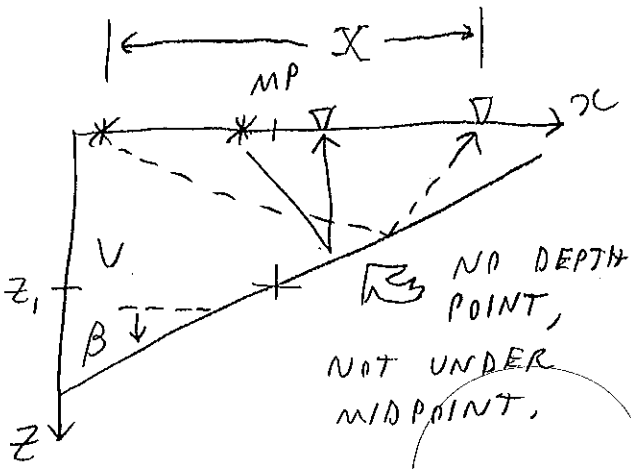
$$\boxed{t_0 = 2\frac{Z}{V}}$$ for zero source-receiver offset X

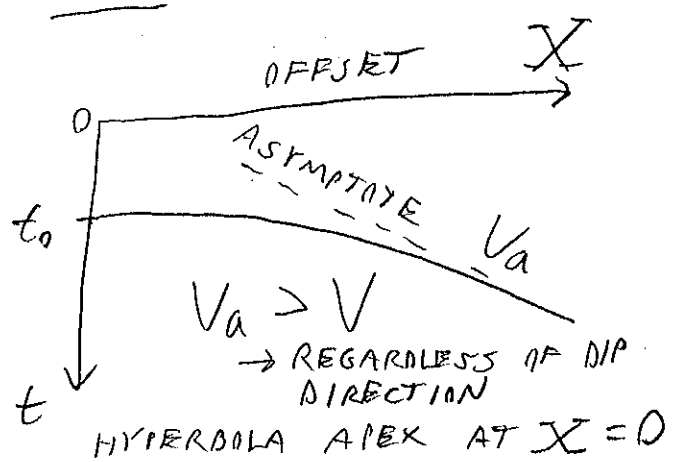
And:

$$\boxed{t^2 = t_0^2 + 4\frac{X^2}{V^2}}$$ The NMO Equation

– We use this hyperbolic equation to find the depths of layers and the velocities above them from the shape of the "normal" moveout.

DIPPING REFLECTORS – Why we have to both profile and sound for reflectors.



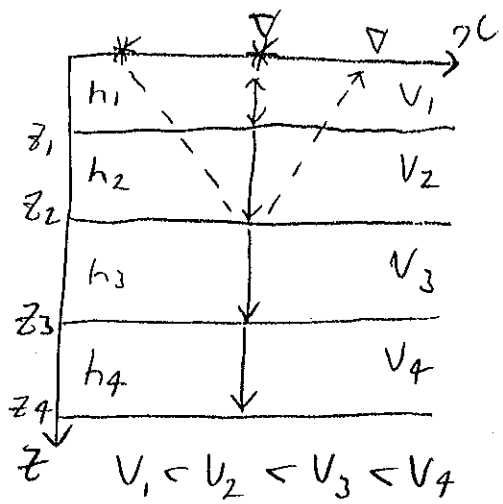


$$\boxed{t^2 = t_0^2 + 4\frac{X^2\cos^2\beta}{V^2}}$$

$$\boxed{V_a = \frac{V}{\cos\beta}}$$

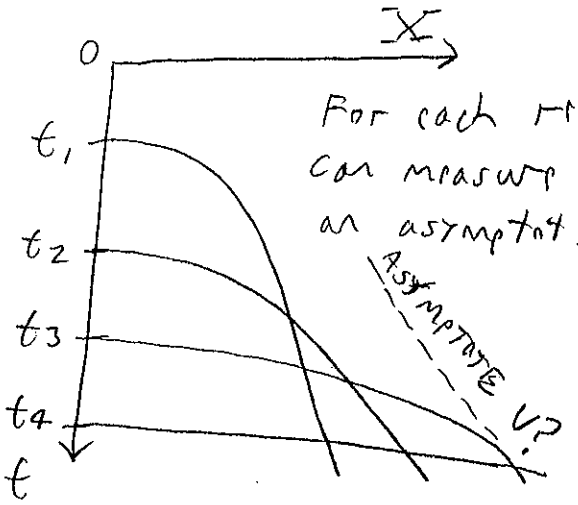
LAYERED FLAT REFLECTORS

FIRST CUTS AT reflection processing and interpretation assume no dip, then let <u>migration</u> take care of that later.



$V_1 < V_2 < V_3 < V_4$

(NOT NECESSARY)



For each reflector we can measure a <u>t_0</u> and an asymptotic <u>velocity</u>.

What will the asymptotic, <u>NMO velocity</u> be for a reflector below a stack of layers having different velocities? It is approximated by the root-mean-squared velocity:

$$\boxed{V_{RMS}^2 = \frac{\sum_{i=1}^{n} V_i^2 \Delta t_i}{\sum_{i=1}^{n} \Delta t_i}} \approx V_{NMO}^2 \text{ of } n^{TH} \text{ reflector down}$$

V_i = real (<u>interval</u>) velocity of i^{TH} layer down

$\Delta t_i = \frac{h_i}{V_i}$ = one-way vertical travel time through i^{TH} layer down.

<u>Note</u> that t_1, t_2, etc. above are two-way travel times, so:

$$\Delta t_i = \frac{1}{2}(t_i - t_{i-1})$$

Getting the Interval (Real) Velocity

C. Hewitt Dix (1955) showed that, as long as the maximum X you use to measure the curvature of the hyperbolas (and derive the asymptotic velocity) is small compared to reflector depths, you can get the layer interval velocities:

$$V_i^2 = \frac{V_{i\,RMS}^2\, t_i - V_{(i-1)\,RMS}^2\, t_{i-1}}{t_i - t_{i-1}}$$

t_i = 2-way vertical time of reflection below layer

$V_{i\,RMS}$ = asympt. velocity of reflection below layer

t_{i-1} = time of reflection above layer

$V_{(i-1)\,RMS}$ = asympt. vel. of reflection above layer

V_i = real (interval) velocity of i^{TH} layer down

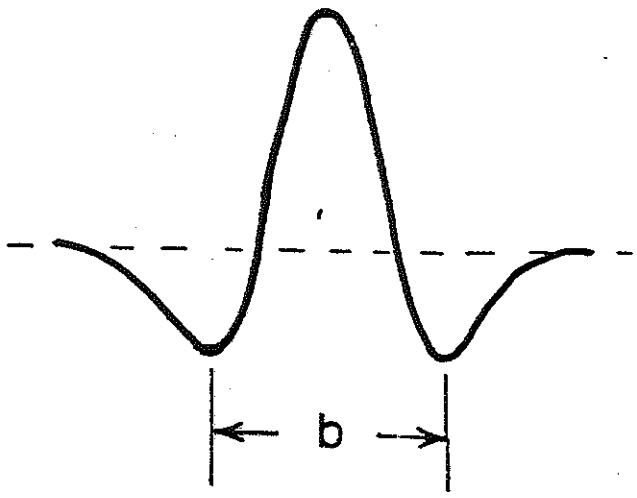
→ If we can get a good measure of V_{RMS} on two reflections, we can get the velocity of the layer in-between, even if it is a Low-Velocity Layer.

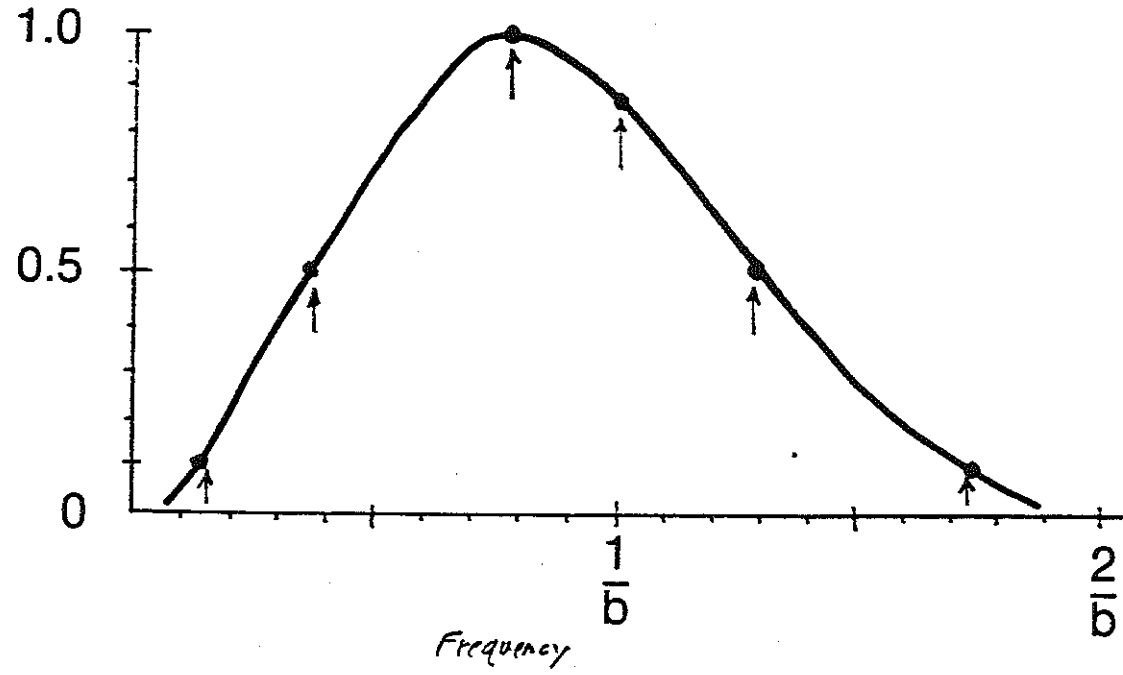
Getting depth

The thickness of the i^{TH} layer down is:

$$h_i = \frac{V_i}{2}(t_i - t_{i-1})$$

Ricker Wavelet





b = Wavelet Breadth

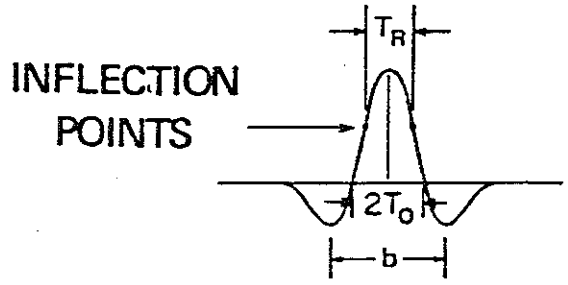
$\frac{0.780}{b}$ = Peak frequency (f_p)

$\frac{0.375}{b}$ & $\frac{1.276}{b}$ = 6dB down (0.5)

$\frac{0.152}{b}$ & $\frac{1.725}{b}$ = 20dB down (0.1)

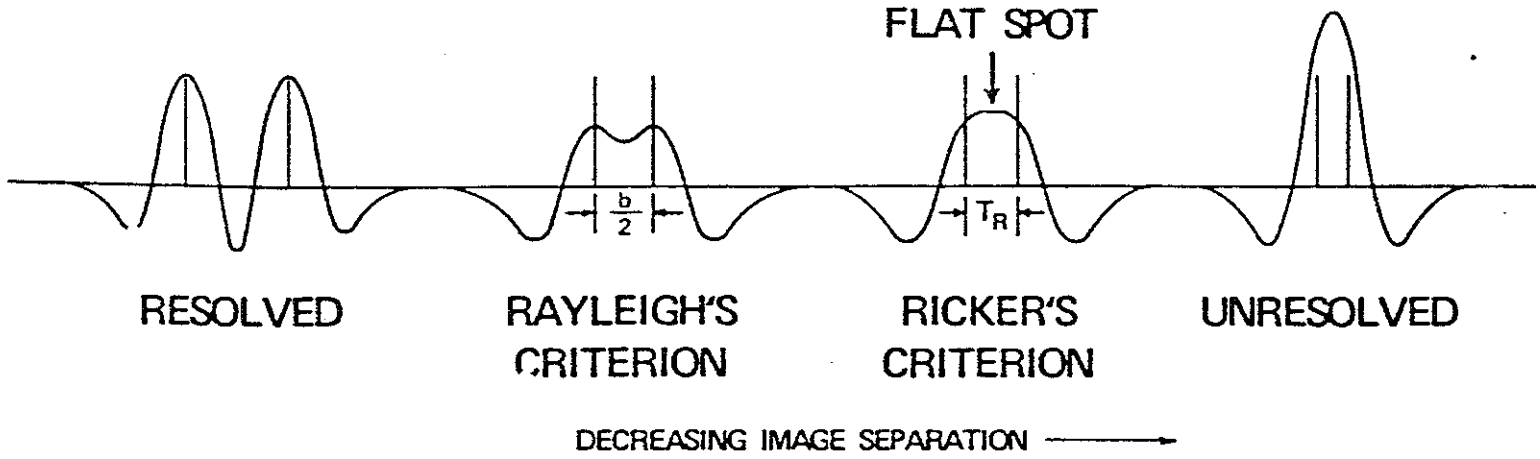
$\frac{1}{b}$ = Predominate Frequency

Vertical Resolution



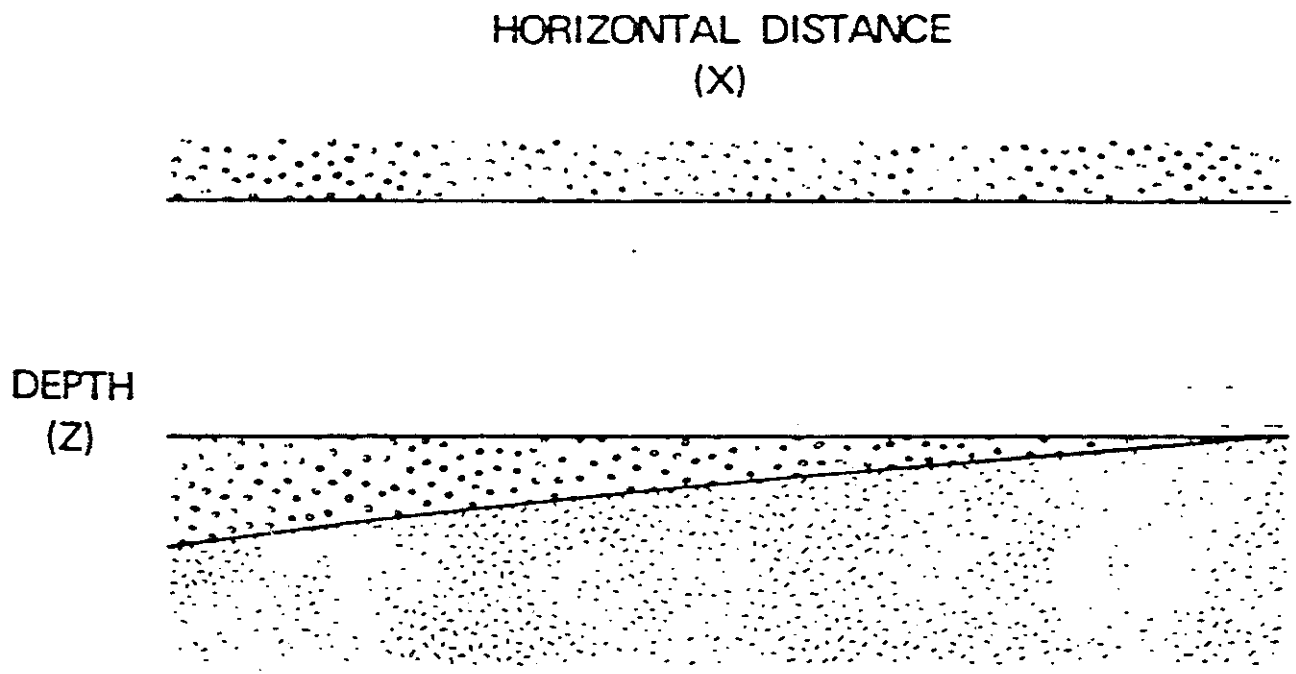
WAVELET

b = WAVELET BREADTH
b/2 = PEAK - TO - TROUGH TIME
$2T_0$ = FIRST ZERO CROSSING INTERVAL



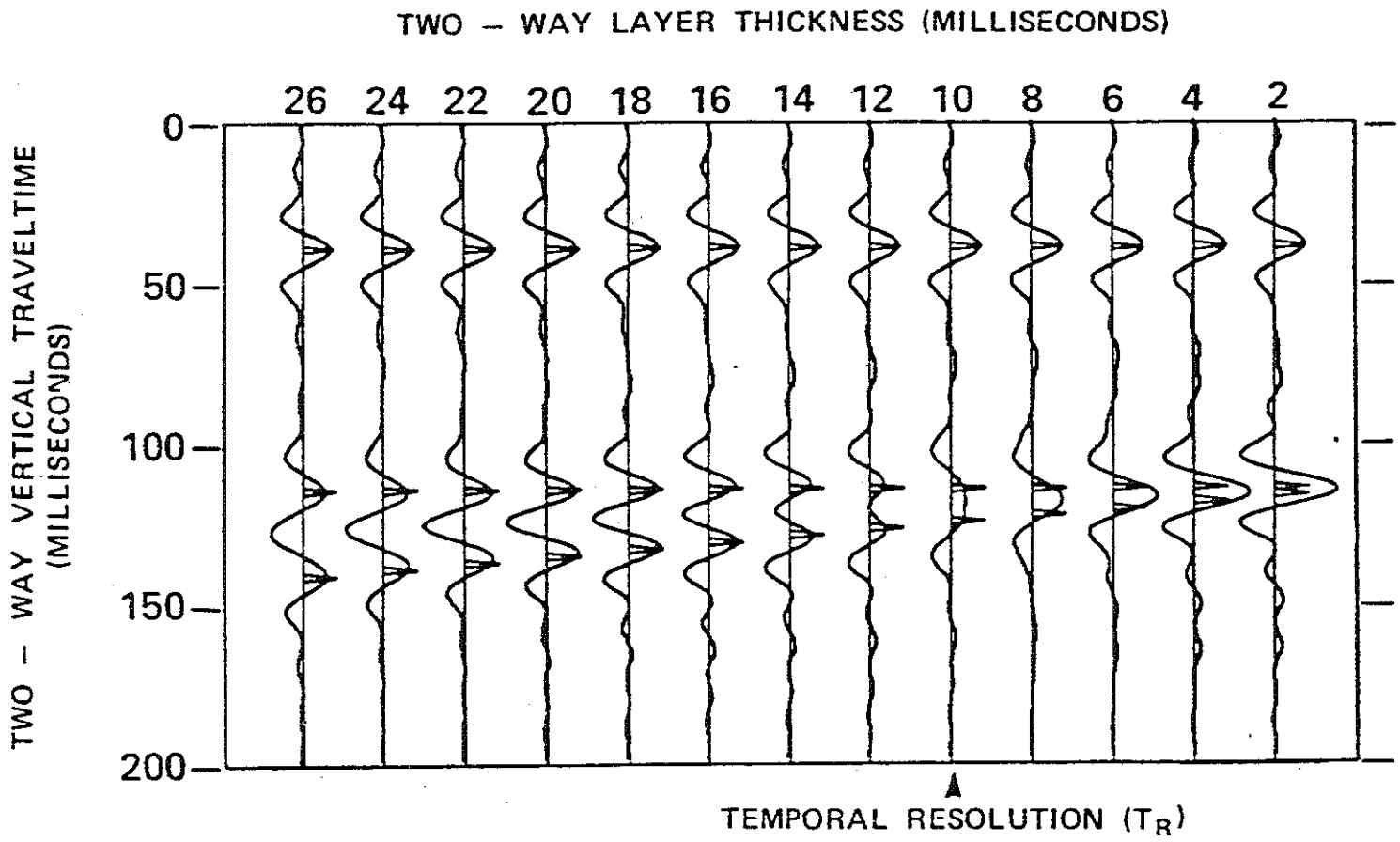
Rayleigh's limit of resolution occurs when images are separated by the peak-to-trough time interval, whereas Ricker's limit occurs when they are separated by a time interval equal to the separation between inflection points.

KALLWEIT and WOOD



The basis for two-term reflectivity spike models consists of a wedge of material bounded above and below by dissimilar material.

KALLWEIT and WOOD



The seismic response of a wavelet convolved with two spikes of equal amplitude and equal polarity is a composite wavelet that varies as a function of spike separation, i.e., layer thickness.

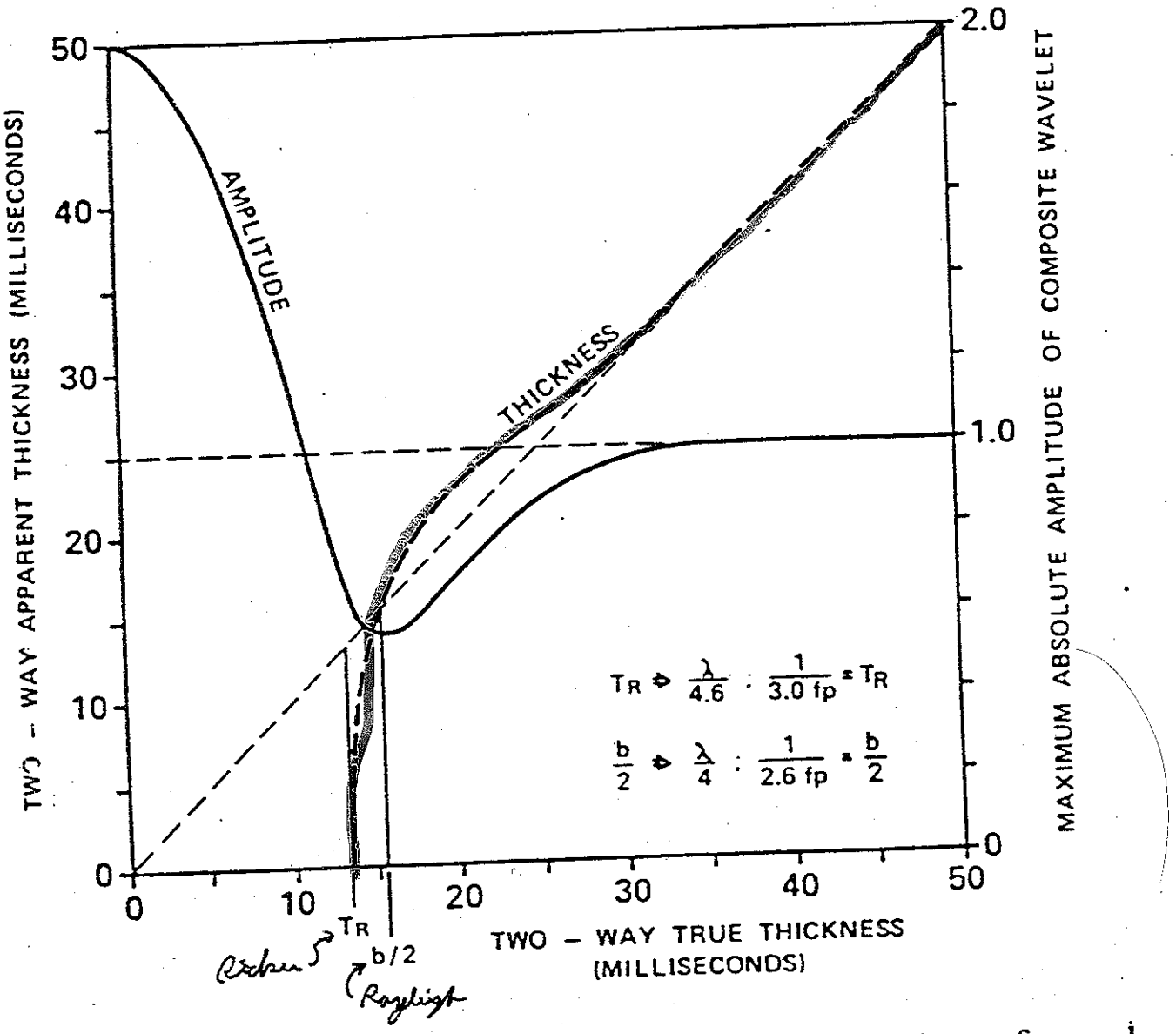
KALLWEIT and WOOD

Vertical Resolution

Widess	= $\lambda/4$	(dominant frequency)
Ricker	= $\lambda/4$	(dominant frequency)
Ricker	= $\lambda/5.2$	(peak frequency)
Kallweit Wood	= $\lambda/4.5$	(mid-frequency of two octave bandwidth)
Kallweit Wood	= $\lambda/2.8$	(upper frequency of two octave bandwidth)

wavelength = velocity/frequency

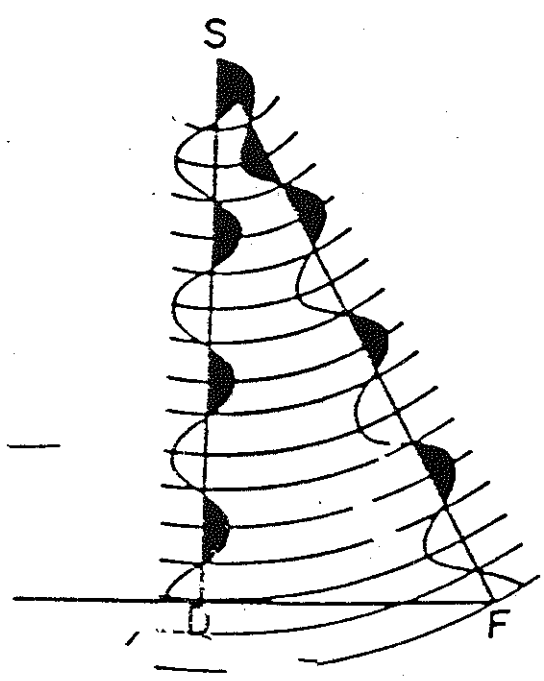
$V = f\lambda \qquad \lambda = \frac{V}{f}$

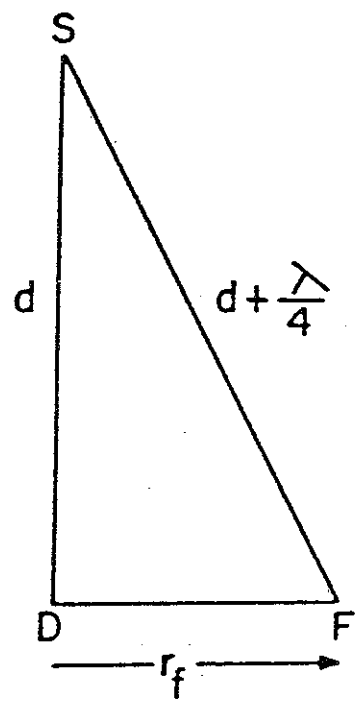


Resolution and detection graphs for two spikes of equal amplitude and equal polarity convolved with a 25-Hz Ricker wavelet show apparent thickness (peak-to-peak time) and maximum absolute amplitude of the composite wavelet as a function of true thickness (spike separation).

KALLWEIT and WOOD

Horizontal Resolution





$$r_f \approx \frac{V}{2}\sqrt{\frac{t}{f}}$$

r_f = radius of Fresnel zone

V = velocity

t = two way time

f = dominate frequency

TATHAM AND MASSELL